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Racial disparities in the Paycheck Protection Program ☆

Sergey Chernenko [a,*], David Scharfstein [b]

[a] Purdue University, 403 W. State Street, West Lafayette, IN 47907, United States of America
[b] Harvard Business School, Soldiers Field, Boston, MA 02163, United States of America

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ABSTRACT

Consistent with contemporaneous research, we document that minority-owned firms were more likely than observationally similar white-owned firms to receive PPP loans from nonbank lenders than from banks. However, we show that this substitution to nonbanks was only partial, resulting in significantly lower PPP take-up by minority-owned firms, particularly Black-owned ones. Location and firm characteristics explain about two-thirds of the 25 percentage point disparity in PPP take-up by Black-owned firms. While there was greater substitution to nonbanks in more racially biased locations, overall take-up was still lower in those locations. Access to professional help with applications facilitated use of nonbanks and mitigated disparities.

1. Introduction

The Paycheck Protection Program (PPP), authorized by the CARES Act in March 2020, was a key component of the U.S. government's response to the adverse economic effects of the COVID-19 pandemic. Under the PPP, the Small Business Administration (SBA) guaranteed about $800 billion in low-interest loans made by financial institutions to businesses with up to 500 employees, promising to forgive these loans if borrowers maintained employment and certain other fixed expenses. A number of concerns have been raised about the design of this program, including whether it targeted firms most needing support, whether it was subject to considerable amounts of fraud, and whether it provided equal access to minority-owned firms.

This paper provides the first systematic micro-level analysis of the take-up of PPP loans by minority-owned businesses. We document significant disparities in PPP borrowing and explore the role of location, firm characteristics, borrowing relationships, access to CPAs, and racial bias in explaining these disparities. We do so by studying Florida restaurants, exploiting state administrative data on restaurant licenses, corporate records, voter registration, lien filings, and mortgage records, as well as detailed information on restaurants from Yelp. Restaurant license data give us the population of restaurants in Florida, almost all of which should have been eligible to receive PPP loans. From corporate records, we determine the identity of the restaurant owner. Voter registration data tell us the restaurant owner's self-identified racial and Hispanic identity. Data on firms' existing secured business loans and on home mortgages taken out by the firm's owners enable us to examine the effect of prior borrowing relationships on PPP take-up. Detailed data on restaurant characteristics from state licenses and Yelp allow us to control for differences among restaurants that could affect PPP loan supply and demand.

To help us understand racial disparities in the PPP, we also contrast the PPP with the Economic Injury Disaster Loan (EIDL) program, an existing SBA program that was significantly expanded to mitigate the adverse economic impact of the pandemic. As part of this program, the SBA made non-forgivable low-interest loans directly to small businesses without using financial intermediaries. It also made grants up to $10,000 under the EIDL Advance program. Studying the EIDL program in combination with the PPP allows us to control for differences in firm and owner characteristics that might affect the demand for emergency

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* Corresponding author.
E-mail addresses: schernen@purdue.edu (S. Chernenko), dscharfstein@hbs.edu (D. Scharfstein).

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support. It also helps us expand our analysis to other industries by identifying firms that were likely eligible for PPP loans.

We start by documenting that Black-owned restaurants are 25.5 percentage points less likely than white-owned restaurants to receive PPP loans. The difference is 10.7 percentage points for Hispanic-owned restaurants and 2.3 percentage points for Asian-owned restaurants. Female-owned restaurants are 4.8 percentage points less likely to receive PPP loans. All of these disparities are statistically significant at 1%, except for Asian-owned restaurants, which experience a disparity that is statistically significant at 10%.

The disparities in overall PPP borrowing are driven by disparities in bank borrowing. Black-owned restaurants are 34.6 percentage points less likely than white-owned restaurants to receive PPP loans from banks, while Hispanic-owned restaurants and Asian-owned restaurants are 11.3 and 10.0 percentage points less likely to receive PPP loans from banks. The difference for female-owned restaurants is 5.9 percentage points. Except for Hispanic-owned restaurants, these disparities in bank borrowing tend to be offset by greater borrowing from nonbank PPP lenders, largely fintechs. While the substitution is large enough to eliminate disparities in PPP borrowing for Asian-owned restaurants, this is not the case for Black-, Hispanic- and female-owned restaurants, which are still less likely than white-owned restaurants to receive PPP funding. Moreover, the racial disparities we document in the PPP do not exist in the EIDL program; if anything, minority-owned restaurants, particularly Hispanic-owned ones, are more likely to receive EIDL loans. As noted, these loans are not intermediated by banks, but rather are made directly by the SBA.

What explains these disparities in PPP take-up, particularly the much lower rate of PPP borrowing by Black-owned restaurants? We consider six potential explanations: (1) location; (2) firm characteristics; (3) other sources of differences in demand for emergency support; (4) pre-existing borrowing relationships; (5) access to professional help with PPP applications; and (6) racial bias.

The first explanation, location, accounts for about 10%–25% of the disparity in PPP borrowing by minority-owned businesses. Once we include ZIP code fixed effects in our regressions, estimated disparities in PPP borrowing fall from 25.5 to 19.0 percentage points for Black-owned businesses and from 10.7 to 8.9 percentage points for Hispanic-owned businesses. About two-thirds of the ZIP code fixed effect can be explained by two ZIP code characteristics: bank branches per capita and median household income. Restaurants in ZIP codes with fewer bank branches per capita and lower household income are less likely to get PPP loans.

We examine the second explanation, firm characteristics, by controlling for restaurant size, age, and a variety of other characteristics derived from Yelp data, including credit card acceptance and the number of reviews and photos. Older, larger, and more heavily visited and reviewed restaurants are more likely to receive PPP funding. Controlling for these characteristics reduces the disparity for Black-owned restaurants by another 10 percentage points, or 40% of the unconditional disparity. However, Black-owned restaurants are still 9.2 percentage points less likely to receive PPP loans. For Hispanic-owned restaurants, the disparity in PPP borrowing is cut by a similar percentage with the inclusion of these firm characteristics, resulting in a 5.7 percentage points disparity in PPP borrowing.

We next explore whether the estimated disparities in PPP take-up could be explained by lower demand for PPP loans by minority-owned restaurants, even though our extensive list of firm characteristics and ZIP code fixed effects are meant to soak up most of the variation in demand. On its face, this explanation of disparities is unlikely to have much traction given that minority-owned restaurants substituted from bank PPP loans to nonbank PPP loans and EIDL loans. Nevertheless, minority-owned restaurants may have been more vulnerable to the pandemic and thus more likely to shut down and not apply for PPP loans. However, we do not find that this is the case; controlling for observable characteristics, minority-owned restaurants are no more likely to shut down than white-owned restaurants. Moreover, our findings on a restricted sample of surviving restaurants are similar to our findings in the full sample. Controlling for restaurant visits during the pandemic also does not affect the estimated racial disparities in PPP borrowing.

We also address the concern that racial disparities are driven by lower demand for PPP loans from minority-owned restaurants by examining PPP take-up in a sample of firms that received grants of $1,000 per employee, up to a total of 10 employees, as part of the EIDL Advance program. Focusing on these grant recipients allows us to analyze PPP borrowing in a sample of firms with a demonstrated awareness of government emergency support programs and demand for such support. It also allows us to directly control for the number of employees. We find that among firms that receive EIDL Advance grants, Black-owned businesses are significantly less likely to receive PPP loans than white-owned businesses with the same number of employees.

Having ruled out differences in demand as an explanation of conditional disparities in PPP take-up, we next investigate the role of bank relationships, access to professional help with PPP applications, and racial bias. Existing bank relationships may be associated with higher take-up of PPP loans for a number of reasons. During the first round of the PPP, banks may have prioritized their existing clients because of limited funds and capacity to process applications (Bartik et al., 2023; Balyuk et al., 2021; Li and Strahan, 2021). Although funding and processing constraints were significantly relaxed in the second round, banks still had to trade off the costs and benefits of processing PPP applications. Some banks and loan officers may have felt that the cost of helping small businesses with weaker documentation and without access to professional help was high compared to the income generated by successfully approved loans. Existing relationships, especially those with credit exposures, would create an additional incentive for a bank to help a small business navigate the PPP application process. Bank borrowing relationships may also proxy for a firm's financial sophistication, record keeping, and access to CPAs and other professional service providers.

While we cannot measure an entire banking relationship, including different types of loans, checking accounts, credit cards, and merchant services, we use data on Uniform Commercial Code (UCC) filings to determine whether a firm had outstanding secured loans.[1] These loans likely capture the main variation across restaurants in their use of banking services, as almost all of the restaurants in our sample accept credit cards and thus also use business checking accounts and merchant services. We also use mortgage records to identify firms whose owners had taken out home mortgages. We find that while minority-owned restaurants are much less likely to have outstanding secured bank loans or to have taken out home mortgages from banks, controlling for prior bank borrowing does little to reduce the measured disparities in bank PPP borrowing.

We next study whether disparities in access to professional services — which many firms used to aid them in filling out their applications — can help explain disparities in PPP take-up. To analyze this possibility, we construct measures of access to Certified Public Accountants (CPAs). The first measure is an indicator for whether a firm uses a CPA as its registered agent in its corporate registration with the state.[2] Our second measure is the number of CPAs in the firm's ZIP code, which tries to capture the available supply of providers that could have helped small firms with their PPP applications. We separately calculate the total number of such CPAs as well as the number of Black CPAs, and we interact these measures of access to CPAs with the minority-owner dummies. We find that Black-owned restaurants benefit significantly more from having a CPA as their registered agent and from having Black CPAs in their ZIP codes. Black-owned firms with a CPA registered

[1] Balyuk et al. (2021) also use UCC filings to identify bank borrowing relationships and ask whether banks prioritized relationship borrowers when processing PPP applications.

[2] A registered agent is a business or individual designated to receive notice of legal or administrative actions affecting the firm. A firm's registered agent may be one of its officers or a third party such as a lawyer or CPA.

agent are just as likely to receive a PPP loan as are observably similar white-owned firms. Having access to an additional Black CPA in a ZIP code is associated with a 8.1 percentage point smaller disparity in the take-up of PPP loans. This means that in ZIP codes with two Black CPAs, there is no disparity in the take-up of PPP loans.[3] These results on both CPA registered agents and the number Black CPAs in a ZIP code are driven by an increase in the take-up of nonbank PPP loans. The findings suggest that CPAs may be particularly helpful in navigating the application process at nonbanks, which generally ran leaner operations and likely did not have the resources to support firms through the application process. The administrative burdens of the PPP application process — and their implications for disparities in application and approval rates — are discussed and documented in detail in Chernenko et al. (2024)

Finally, we explore how racial bias may have affected both the overall take-up of PPP loans as well as their source, i.e. bank versus nonbank lenders. It is possible that there was racial bias in the way banks processed PPP loan applications. Indeed, audit studies by the National Community Reinvestment Coalition (NCRC) find that Black, Hispanic, and female business owners received worse treatment from banks in response to inquiries about PPP loans (National Community Reinvestment Coalition, 2020b,a). It is also possible that even if there was no discrimination in the application process at banks, a legacy of past discrimination and poor treatment discouraged minority-owned businesses from even approaching banks for a PPP loan.

We use measures of implicit and explicit racial bias from Project Implicit, which offers online tests to measure a person's implicit associations and biases.[4] Our findings indicate that Black-owned restaurants in counties where white test takers exhibit more explicit and implicit bias towards Black people are significantly less likely to receive PPP funding from banks. The effect is large: a one standard deviation increase in implicit bias is associated with a 19.1 percentage point reduction in the probability that a Black-owned restaurant receives a PPP loan from a bank. Moreover, in more racially biased counties, Black-owned businesses are much more likely to substitute to nonbank PPP loans and EIDL loans from the SBA, though the former effect is statistically significant only when using implicit bias as our measure. Because applications to nonbanks and the SBA are typically made online, while applications to banks often involve more personal interaction, there may have been less scope for racial bias in the application process at nonbanks and the SBA. This may help explain why Black business owners are more likely to use these funding sources.

Finally, we broaden our analysis to other industries and other states. While we do not have the full population of eligible firms in other industries to study PPP take-up, we can study disparities in PPP borrowing among the sample of firms that received EIDL Advance grants. Studying this sample has the further benefit of allowing us to expand our sample to include firms in Georgia, Louisiana, and North Carolina, other states with publicly available voter registration data that includes self-reported race. With this larger sample, we can separately control for the interaction between Black business ownership and county characteristics that may be correlated with racial bias. We can also ask whether minority-owned businesses benefit more from existing bank borrowing relationships.[5] This may be the case if minority-owned firms are subject to greater administrative burdens or existing bank borrowing relationships help firms overcome the effects of racial bias. In regressions with ZIP code cross industry fixed effects and controls for firm age, number of employees, and secured borrowing relationships, Black- and Hispanic-owned businesses are significantly less likely to receive PPP loans. The magnitudes of racial disparities are broadly similar to our sample of Florida restaurants. Furthermore, Black-owned businesses are even less likely to receive PPP loans in more racially biased counties. Finally, the effect of existing borrowing relationships is significantly stronger for Black- than for white-owned businesses, suggesting that existing borrowing relationships help mitigate disparities in PPP take-up but do not completely eliminate them.

Our paper is part of a now sizable literature studying the functioning and impact of the PPP,[6] including a number of recent studies on racial disparities in the PPP. Wang and Zhang (2020) show that ZIP codes with a greater percentage of Black residents had less PPP take-up, measured by the ratio of PPP loans to the number of establishments in a ZIP code. They find that less PPP take-up is related to a lower concentration of branches of PPP-approved lenders in ZIP codes with more Black residents. Consistent with this finding, Erel and Liebersohn (2022) document that fintech lenders originated a larger share of PPP loans in ZIP codes with a larger minority population share. While these findings are important in that they suggest that location matters in accessing PPP funding, our firm-level approach allows us to look beyond location and examine the role of firm characteristics, borrowing relationships, access to professional help, and racial bias. Indeed, we show that ZIP code fixed effects account for only one-fifth of racial disparities in PPP borrowing, while firm characteristics, access to professional help, and racial bias have a much larger collective impact.

In a contemporaneous paper analyzing the full population of PPP borrowers, Howell et al. (2024) show that Black-owned businesses that receive PPP funding are less likely to get funded by small banks than they are to get funded by either fintechs or the four largest banks. In a similar vein, Fei (2022) use the sample of PPP borrowers matched to Yelp-listed restaurants to show that minority-owned restaurants are more likely to get their PPP loans from fintech lenders rather than from banks. Both Howell et al. (2024) and our paper find that Black-owned businesses are more likely to borrow from fintechs in more racially biased locations and that controlling for prior banking relationships does not meaningfully reduce average disparities in bank PPP funding. All three papers suggest that online loan application processes attenuate racial disparities in PPP take-up. Howell et al. (2024) show that adoption of automated processes by banks is associated with an increase in the share of their PPP lending to Black-owned businesses, suggesting that automation could reduce racial disparities.

There are at least four distinctive features of our paper. First, and most importantly, our focus on restaurants and then EIDL Advance grant recipients allows us to analyze a set of firms that were likely to qualify for the PPP. We are thus able to examine racial disparities in PPP access, not just whether minority-owned firms are more likely to get their PPP funding from banks or nonbanks. The prior literature has been mostly silent on whether there are racial disparities in firm-level PPP take-up. Our methodology allows us to conclude that even though Black-owned businesses rely more on nonbank lenders, this substitution is not enough to offset the disparity in bank PPP funding. Our analysis further suggests that racial bias has real effects on PPP take-up.

Second, we provide novel evidence that lack of access to professional service providers such as CPAs is associated with significantly larger disparities in PPP take-up. Our results point to an important limitation of nonbank lenders in expanding access and reducing racial disparities in access to credit. Because nonbanks — most of them fintechs — have limited capacity to help firms fix any deficiencies in their applications, fintechs work best when minority-owned businesses have access to professional service providers.

Third, we examine PPP take-up in conjunction with the EIDL program, the other significant emergency support program for small businesses, which was expanded during the pandemic. We show that there

[3] The average in our sample is 0.24.

[4] https://implicit.harvard.edu/implicit/takeatest.html.

[5] The restaurants sample has too few Black-owned firms with bank UCC loans to test for differential effects of bank borrowing relationships.

[6] See, for example, Autor et al. (2022), Bartik et al. (2023), Granja et al. (2022), Li and Strahan (2021), Hubbard and Strain (2021) and Griffin et al. (2023).

is greater take-up of EIDL loans among minority-owned firms, despite the less attractive terms of this program relative to the PPP. Minority-owned firms may have been more likely to use EIDL loans because applications were submitted online directly through the SBA, and like fintech PPP loans, were not intermediated by banks.

Fourth, by linking business owners to voter registration data, we have a better measure of self-identified race and Hispanic origin than machine-learning algorithms commonly used to infer race from individual characteristics, such as an individual's name and location. As we show in Chernenko and Scharfstein (2023), measurement error in predicted race can significantly affect the magnitude of the estimated racial disparities as well as the conclusions one draws about the sources of these disparities. Predicted race tends to underestimate unconditional disparities, overstate the significance of location, and significantly underestimate conditional disparities. Predicted race can also affect one's conclusions regarding interaction terms, i.e., cross-sectional variation in disparities.

In a more recent paper, Chernenko et al. (2024) use the 2020 Small Business Credit Survey conducted by the Federal Reserve Banks to study the relative importance of disparities in loan applications and approvals in explaining racial disparities in the PPP. Using a very different sample and set of controls, Chernenko et al. (2024) find remarkably similar unconditional and conditional disparities in the take-up of PPP loans. They show that racial bias affects both application behavior and loan approvals, and they provide evidence that residual disparities in the take-up of PPP loans are likely due to disparities in the administrative burdens experienced by Black-owned firms in determining eligible loan amounts and providing documentation for the requested loan amounts. Our finding that disparities are mitigated when Black-owned firms have greater access to CPAs is consistent with the explanation put forward in Chernenko et al. (2024).

In addition to contributing to the literature on the PPP, our paper contributes to a broader literature on discrimination in small business lending. Blanchflower et al. (2003), Cavalluzzo and Wolken (2005), Blanchard et al. (2008), and Fairlie et al. (2022) use survey data to show that minority-owned businesses are more likely than white-owned businesses to be turned down for bank loans and are less likely to apply for loans for fear of being turned down. Fairlie et al. (2022) show that these effects are stronger in locations with greater racial bias, consistent with our findings. These studies suggest that our findings could, in part, be driven by a historical legacy of discrimination that discouraged Black-owned businesses from applying for PPP loans from banks.

The rest of the paper is organized as follows. Section 2 describes our data and reports basic summary statistics. Section 3 documents the existence of racial disparities in PPP borrowing, showing that they are driven by disparities in PPP borrowing from banks and that there is only a partial substitution to nonbank sources of PPP funding. We also document the role of location and restaurant characteristics in explaining racial disparities in PPP borrowing. Section 4 provides evidence against the view that racial disparities in PPP borrowing can be explained by lower demand from minority-owned businesses for emergency loans. We examine the role of bank relationships, access to CPAs, and racial bias in Section 5. Section 6 broadens our analysis to other industries and states by studying PPP take-up by EIDL Advance recipients in Florida, Georgia, Louisiana, and North Carolina, states with publicly available voting records that identify racial and Hispanic identity. Section 7 concludes.

2. Data

Our main data set is composed of restaurants in Florida with information on the owner's racial and Hispanic identity. We construct the main sample by combining information from the following data sets: (1) Florida restaurant licenses; (2) Florida corporate records; (3) Florida voter registration; and (4) Yelp data on restaurant characteristics and activity. We then determine whether each restaurant in the sample received loans from the Paycheck Protection Program (PPP) or the COVID-19 Economic Injury Disaster Loan (EIDL) program, both sponsored by the SBA. The EIDL program offers long-term, low-interest loans to firms that are adversely affected by a disaster such as the COVID pandemic. Unlike PPP loans, EIDL loans are not forgivable, and firms apply directly to the SBA for approval, not through an intermediary. Details on the sample construction and the data are described below.

It is worth noting that because our sample consists of licensed restaurants that are matched to corporate records and to Yelp, it is highly unlikely that our results could be explained by the types of fraud documented by Griffin et al. (2023).

2.1. PPP and EIDL loans

Data on approved PPP and EIDL loans are from the SBA website.[7] PPP loan data include all loans approved during the periods April 3–August 9, 2020, and January 11–June 30, 2021. EIDL data include loans approved through November 14, 2020. The SBA has not yet released loan-level data on the EIDL loans made after this date. In 2020, the SBA made 3.6 million EIDL loans for a total of $194 billion, while in 2021 the SBA made about 3.8 million EIDL loans for a total of $284 billion. However, most of our analyses compare PPP and EIDL loans extended during 2020. Both PPP and EIDL data report the borrower's name and location, loan amount, and approval date. PPP loan data also report the lender's name and location, the borrower's industry, and self-reported demographic information on the borrower. The vast majority of PPP borrowers (83%) do not report information on their racial and Hispanic identity.

After limiting the sample to borrowers located in Florida and excluding non-profit organizations, we match borrowers by name to Florida corporate records. Details of the matching algorithm are described in the Internet Appendix. We are able to identify 87.9% of all PPP borrowers and 84.5% of all EIDL borrowers. Most of the unmatched borrowers are individuals.

2.2. Potential borrowers

Much of the existing research on the PPP is constrained by the limited data on small private firms, which comprise most eligible borrowers. We overcome this limitation by studying the take-up of PPP and EIDL loans by Florida restaurants.[8] Because essentially all restaurants were eligible for PPP and EIDL loans and because Florida restaurants are subject to state licensing, we have comprehensive and reliable data on the population of eligible firms. We obtain the list of all Florida restaurant licenses from the Florida Department of Business & Professional Regulation.[9]

To generate a relatively homogeneous sample, we focus on restaurants that offer seating, while excluding food trucks, takeout-only restaurants, and caterers. We also exclude restaurants with licenses approved after February 15, 2020, because they would not have been eligible for the PPP. Finally, we exclude hotel restaurants and franchise restaurants because they are frequently owned and operated by affiliated entities. As a result, it can be difficult to determine whether a given hotel or franchise restaurant received an emergency loan or benefited from a parent receiving an emergency loan.[10] Restaurants are classified

[7] PPP loan data are available at https://sba.app.box.com/s/5myd1nxutoq8wxecx2562baruz774si6, EIDL loans data are available at https://data.sba.gov/dataset/covid-19-eidl.

[8] Restaurants, defined as borrowers with NAICS codes starting with 722, account for 6.4% of all PPP loans.

[9] http://www.myfloridalicense.com/DBPR/hotels-restaurants/public-records/.

[10] In the robustness analyses in the Internet Appendix Table IA1, we use an algorithm to identify affiliated firms and find larger disparities in bank PPP borrowing when we measure borrowing across all of the restaurant's affiliates.

as being in a hotel if they share the same address as one of the hotels in Florida hotel license data.[11] We classify restaurants as franchises based on the restaurant name appearing in the SBA franchise directory.[12]

Restaurant license data identify the name and location of the restaurant as well as the license holder. We match license holders to Florida corporate records. Most license holders are firms, enabling a straightforward match based on name and location. In some cases, the license holder is an individual. We attempt to match these individuals to Florida corporate records based on the name of the firm's first listed officer or director. Because a person can serve as an officer or director of multiple firms, we consider only unique matches, i.e., cases where the restaurant license holder is an officer or director of only one firm in Florida corporate records. Overall, we can match 91% of restaurants that meet the sample selection criteria. Most of the remaining restaurants list individuals as license holders and cannot be unambiguously matched to corporate records.

Table A.2 summarizes the sample selection criteria and the number of restaurants matched to Florida corporate records. We exclude from the analysis restaurants owned by non-profits, out-of-state firms,[13] and publicly-traded firms. We also exclude restaurants matched to firms registered for the first time after February 15, 2020.

2.3. Racial and Hispanic identity of potential borrowers

We classify firms based on the first officer or director listed in the firm's corporate record whose racial and Hispanic identity we can ascertain.[14] For brevity, we refer to this individual as the owner, although it is possible that this individual manages the firm without having any ownership.

Howell et al. (2024) use a machine learning algorithm to predict racial and Hispanic identity in the PPP sample based on the officer's name and the location of the business. Regression results that use predicted racial and Hispanic identity must be interpreted with caution because measurement error will cause the coefficients on the predicted identity to understate the true effect of racial and Hispanic identity, with the magnitude of the bias depending on the relative accuracy with which the algorithm classifies different groups.

In addition to introducing biases in the estimates of the direct effect of racial and Hispanic identity, name-based algorithms could introduce biases in the indirect effects of racial and Hispanic identity to the extent that the measurement error is correlated with firm and location characteristics. Fryer and Levitt (2008) show that the use of distinctly Black names varies over time and is strongly correlated with socioeconomic status. Thus, for example, if there is less measurement error in more racially biased locations, it is possible that the estimated disparities in PPP take-up in more racially biased locations are overstated because the racial bias measure is really capturing the extent of measurement error in the name-based algorithm. Similarly, if Black-owned businesses are less likely to have existing bank relationships, then measures of bank relationships may provide incremental information relative to a name-based algorithm about whether the business owner is Black.

To overcome the limitations of name-based algorithms, we take advantage of Florida voter registration data, which report each registered voter's self-identified racial and Hispanic identity.[15] Voter registrants can identify themselves as one of the following: (1) American Indian or Alaskan Native; (2) Asian or Pacific Islander; (3) Black, not Hispanic; (4) Hispanic; (5) white, not Hispanic; (6) other; or (7) multi-racial.[16] Because of the small number of observations, we include voters self-identified as American Indian or Alaskan Native or multi-racial in the "other" category. For brevity, we refer to "Asian or Pacific Islander" as Asian, "Black, not Hispanic" as Black, and "white, not Hispanic" as white. These classifications are imperfect, in part because they do not separate race from ethnic identity. Notably, unlike the Census, Florida voter registration only offers "Black" and "Hispanic" as mutually exclusive categories.

We match the firm's owner to voter registration data based on name and location. Details of the matching algorithm are provided in the Internet Appendix. We are able to identify a restaurant owner's racial and Hispanic identity for 70% of restaurants. In most of these cases, we have a unique match within a county. If there are multiple potential matches, but they all report the same racial and Hispanic identity, we use what is reported even though we cannot identify the specific voter match.

One further advantage of using voter registration data is that by conditioning on the owner being a registered voter, we exclude some restaurants that may have been ineligible for PPP funding.[17] The downside is that the sample of registered voters may not be representative of the population of restaurant owners. In particular, registered voters may be more aware of government interventions such as the PPP and thus may be more likely to apply for a PPP loan.

2.4. Restaurant characteristics

Because restaurant license and corporate records data have limited information on restaurant characteristics, we supplement these administrative records data with restaurant characteristics from Yelp, a platform for crowd-sourced information and reviews about restaurants and other businesses. Using name, location, and phone information, we are able to match 89% of restaurants to Yelp. Most of the unmatched restaurant licenses are operated by golf clubs, museums, and other establishments.

Our version of Yelp data includes information on restaurant features, such as whether the restaurant accepts credit cards, offers delivery, or has outside seating. We use features that were in place just before the pandemic. We also have user activity measures, including the number of reviews and photos posted each week, the average rating, and the number of page visits. The cumulative number of reviews since inception and average ratings are as of February 2020. For restaurants without reviews, we set the average rating to 0 and include a dummy for no reviews in our regression analyses. We also include the average number of page views and photos posted each month over the March 2019–February 2020 period.

2.5. Lenders

To classify PPP lenders as either banks or nonbanks, we first match lenders based on name and location to the database on financial institutions maintained by the Federal Financial Institutions Examination Council's National Information Center (NIC). Unmatched lenders and lenders classified by NIC as domestic entity other (DEO) are classified as nonbanks. We also classify Cross River Bank, Celtic Bank Corp, and WebBank as nonbanks because they hold loans originated by online

[11] http://www.myfloridalicense.com/DBPR/hotels-restaurants/lodging-public-records/.

[12] https://www.sba.gov/document/support-sba-franchise-directory.

[13] A firm that is registered in another state but has a Florida mailing address would not be excluded by the earlier screen meant to exclude license holders with out-of-state mailing addresses.

[14] Officers are generally listed in the order of seniority.

[15] https://dos.myflorida.com/elections/data-statistics/voter-registration-statistics/voter-extract-disk-request.

[16] Ganong et al. (2020) show that there is a 99% agreement rate between the way voters identify themselves in a matched sample of voter registration files and mortgage applications.

[17] While the CARES Act does not disqualify foreign-owned businesses from receiving PPP loans, the SBA did not provide explicit guidance, thereby sowing confusion among lenders and causing some lenders to deny applications by non-citizens. One of the changes implemented by President Biden on February 22, 2021 was to require the SBA to provide clear guidance that otherwise eligible businesses cannot be denied access.

Table 1
Summary statistics.

	Main sample ($N = 12,812$)				Corporate records ($N = 20,456$)			
	Mean	SD	25th	75th	Mean	SD	25th	75th
Number of seats	89.66	102.61	35.00	140.00	87.00	102.54	30.00	130.00
Firm age	9.89	9.72	3.15	13.66	9.23	9.60	2.79	12.53
Bank UCC	0.15	0.36	0.00	0.00	0.14	0.35	0.00	0.00
Nonbank UCC	0.14	0.34	0.00	0.00	0.14	0.34	0.00	0.00
Received EIDL	0.36	0.48	0.00	1.00	0.33	0.47	0.00	1.00
PPP	0.70	0.46	0.00	1.00	0.66	0.47	0.00	1.00
Bank PPP	0.64	0.48	0.00	1.00	0.58	0.49	0.00	1.00
Nonbank PPP	0.06	0.24	0.00	0.00	0.07	0.26	0.00	0.00

This table reports summary statistics for the main sample of Florida restaurants and for the larger sample of restaurants not matched to voter registration and Yelp data. The main sample is composed of restaurants with seating that, as of February 15, 2020: (1) were licensed in Florida; (2) were registered as Florida for-profit firms; (3) had an owner whose racial and Hispanic identity can be determined from Florida voter registration data; and (4) can be matched to Yelp. Hotel and franchise restaurants are excluded. The corporate records sample is composed of restaurants that meet the above criteria but are not matched to voter registration data and Yelp data.

fintech lenders and do not originate most of the PPP loans they hold. The list of nonbank lenders is reported in Table A.3. Most nonbank lending is done by fintechs, firms that use more advanced financial technology to process loan applications online.

2.6. Bank relationships

We measure borrowing relationships using Uniform Commercial Code (UCC) filings.[18] Lenders file UCC financing statements to establish priority in the collateral pledged to them. Using Florida UCC filings data,[19] we match debtors to Florida corporate records. We match lenders to the National Information Center (NIC) and Capital IQ, and we classify them as bank or nonbank. For each restaurant in our sample, we then check whether the firm had an active UCC filing and whether the underlying loan was from a bank or a nonbank.

2.7. Summary statistics

Table 1 reports summary statistics for the main sample. For comparison, the table also includes summary statistics for the sample of restaurants matched to corporate records but not matched to voter registration and Yelp. As can be seen from the table, firms in the voter registration sample are quite similar to firms in the full sample based on restaurant seats, age, and whether they have secured loans with banks or nonbanks. They also get PPP loans at roughly similar rates. Thus, restricting the analysis to firms we can match to the voter registration and Yelp samples does not materially affect the types of firms we are studying.

3. Documenting disparities

Table 2 reports sample means for restaurant characteristics, location characteristics, and emergency loan take-up for restaurants broken out by the owner's racial and Hispanic identity and by gender. Minority-owned and female-owned restaurants tend to be younger and smaller. They are also less likely to have an existing secured loan with either a bank or nonbank lender. For example, relative to white-owned restaurants, Black-owned restaurants are only 21% as likely to have a secured loan from a bank (4% vs. 19%). The means for the Yelp variables are fairly similar across demographic groups except for Black-owned restaurants, which have significantly fewer reviews, page views, and photos. The average rating of Black-owned restaurants is similar to that of other groups. Like other demographic groups, almost all Black-owned restaurants accept credit cards.

Minority-owned restaurants tend to be located in ZIP codes with larger populations and lower white population shares. Black-owned restaurants, for example, are located in ZIP codes in which white people comprise 56% of the overall population. In contrast, white-owned restaurants are located in ZIP codes in which white people comprise 81% of the overall population. Black-owned restaurants are also located in ZIP codes with about 45% fewer bank branches per capita, 20% lower median household income, and 17% higher COVID cases per capita. Our regression specifications will account for these geographic differences by comparing white- and minority-owned restaurants located in the same ZIP code.

Table 2 also shows that there are significant differences across groups in the use of emergency loans. Out of white-owned restaurants, 74% receive a PPP loan. Almost all of these are from a bank. A third of white-owned restaurants receive an EIDL loan, with most recipients also receiving a PPP loan. Only 5% of white-owned restaurants receive an EIDL loan without also receiving a PPP loan.

The picture is very different for Black-owned restaurants. Only 48% receive a PPP loan. Conditional on receiving a PPP loan, almost 30% of PPP loans are from a nonbank. Compared to white-owned restaurants, Black-owned restaurants are almost three times more likely to receive only an EIDL loan, and 39% of Black-owned restaurants receive neither type of loan. The pattern of outcomes for other minority-owned restaurants is generally in between those for white- and Black-owned restaurants.

These differences are reflected in Table 3, which reports the results of linear probability model regressions of the various types of emergency loans on indicator variables for minority- and female-owned restaurants. The unit of observation is a restaurant r located in ZIP code z and owned by firm f. A given firm can own multiple restaurants. Table IA1 in the Internet Appendix shows the robustness of our results to analyzing single-restaurant firms, which comprise almost 93% of our sample.

In columns 1–4 of Panel A in Table 3 the dependent variable is equal to one if the restaurant receives a PPP loan and zero otherwise. We include indicators for the racial and Hispanic identities of restaurant owners, while excluding white restaurant-owner indicators, so we are measuring the likelihood that minority-owned restaurants get a PPP loan relative to restaurants with white owners. As shown in the first column, which reports the results for the regression without any controls, Black-owned restaurants are 25.5 percentage points less likely than white-owned restaurants to receive a PPP loan, while Hispanic-owned restaurants are 10.7 percentage points less likely to receive a PPP loan. Both coefficients are statistically significant. Asian-owned restaurants are 2.3 percentage points less likely to receive a PPP loan, although the difference is statistically significant at only 10%. Female-owned restaurants are 4.8 percentage points less likely to receive a PPP loan.

One potential reason why minority-owned restaurants are less likely to receive PPP loans is that they are located in underserved banking markets (Wang and Zhang, 2020). The second column of Table 3 adds bank branch density to the regression. We also include population and median household income in 2019, as well as COVID cases per capita in 2020. All continuous variables are standardized to have a mean of zero and a standard deviation of one.

Restaurants in ZIP codes with greater branch density are more likely to receive PPP loans. A one standard deviation increase in branch density increases the probability of receiving a PPP loan by 2.4 percentage points. Given that only 30% of restaurants do not receive a PPP loan, the increase is meaningful. While the coefficients of log population and of COVID cases per capita are not statistically significant, restaurants in ZIP codes with larger median income are more likely to receive PPP funding. It is possible that restaurants in more affluent locations were

[18] Gopal (2021) and Gopal and Schnabl (2022) also use UCC filings to measure lending to small businesses. These papers provide more details on UCC filings.

[19] https://www.floridaucc.com/uccweb/ucc.aspx.

Table 2
Take-up of emergency loans by different groups.

	Total	Race/Ethnicity				Gender	
		Asian	Black	Hispanic	White	Male	Female
N	12,812	1525	482	2533	7785	8836	3976
Firm characteristics							
Firm age	9.89	7.35	6.91	8.58	11.04	10.02	9.60
Number of seats	89.66	68.72	53.33	73.20	101.86	96.47	74.53
Bank UCC	0.15	0.08	0.04	0.11	0.19	0.17	0.12
Nonbank UCC	0.14	0.07	0.13	0.15	0.15	0.15	0.12
Accepts credit cards	0.99	1.00	0.98	0.99	0.98	0.99	0.98
Number of reviews	128.84	117.56	47.80	119.99	139.23	138.25	107.92
No reviews	0.06	0.04	0.22	0.08	0.05	0.06	0.07
Rating	3.66	3.74	3.06	3.59	3.69	3.66	3.65
Page views	400.99	437.39	237.78	367.46	416.35	430.68	335.02
Photos	2.85	3.47	1.74	2.99	2.76	3.08	2.34
ZIP characteristics							
Population (000)	30.51	33.85	38.33	37.39	26.99	30.37	30.82
White population share	0.78	0.76	0.56	0.77	0.81	0.78	0.78
Bank branches per capita	0.39	0.37	0.23	0.34	0.42	0.40	0.37
Median household income ($000)	60.16	61.78	49.21	57.05	61.64	60.68	59.01
COVID cases per capita	0.07	0.06	0.07	0.09	0.06	0.07	0.07
Outcomes							
Received PPP	0.70	0.71	0.48	0.62	0.74	0.72	0.66
Bank	0.64	0.58	0.33	0.57	0.69	0.66	0.58
Nonbank	0.06	0.13	0.14	0.05	0.05	0.06	0.08
Received EIDL	0.36	0.36	0.36	0.40	0.34	0.36	0.35
EIDL and PPP	0.29	0.30	0.22	0.30	0.29	0.30	0.27
EIDL, not PPP	0.06	0.06	0.14	0.10	0.05	0.06	0.08
No loan	0.24	0.24	0.39	0.27	0.21	0.23	0.26

This table reports sample means broken out by owner's racial and Hispanic identity, and by gender. Population and median household income are in thousands. The main sample is composed of restaurants with seating that, as of February 15, 2020: (1) were licensed in Florida; (2) were registered as Florida for-profit firms; (3) had an owner whose racial and Hispanic identity can be determined from Florida voter registration data; and (4) can be matched to Yelp. Hotel and franchise restaurants are excluded.

hit less hard by the pandemic and thus were more likely to survive and apply for PPP loans. Since Black-owned restaurants tend to be in ZIP codes with lower branch density and lower household income, the estimated magnitude of the disparity in PPP take-up for Black-owned restaurants declines, but the reduction is just 4 percentage points, from 25.5 to 21.4 percentage points. The controls have a similar effect on the estimated disparity for Hispanic-owned restaurants.

Replacing these ZIP code level variables with ZIP code fixed effects in column 3 has a modest incremental effect on the estimated coefficients, reducing the estimated disparity in PPP loans for Black-owned restaurants from 21.4 percentage points with controls to 19.0 percentage points with fixed effects. Thus, a combination of bank branch density, household income, and COVID cases per capita does a good job explaining variation in disparities across ZIP codes. Nevertheless, the disparity for Black-owned restaurants remains large and statistically significant despite the inclusion of ZIP code fixed effects. The estimated disparity for Hispanic-owned restaurants is smaller at 8.9 percentage points but still statistically significant.

We next consider whether restaurant characteristics affect PPP loan demand and supply. Thus, in column 4 we add firm age and size controls, using the log number of restaurant seats as a proxy for size. We also add a variety of restaurant characteristics from Yelp, all measured before the pandemic. Larger and older restaurants are significantly more likely to receive PPP loans, as are those that accept credit cards and have more Yelp customer reviews and posted photos. These variables presumably measure how well-established and popular the restaurant was before the pandemic. There are two reasons to include these controls. The first is that these characteristics could help control for PPP loan demand if smaller, less established, and less popular restaurants were less profitable and thus more likely to shut down during the pandemic and not apply for a PPP loan. The second is that these characteristics could control for loan supply as PPP lenders may have prioritized larger, more established, and more popular restaurants. This could be because they were more likely to be existing customers of the lender. It is also possible that lenders prioritized them because there were more fees from making larger PPP loans or because these restaurants were more likely to survive and become future customers of the lender.

Importantly, controlling for the firm characteristics described above reduces the magnitude of the coefficients on Black- and Hispanic-owned restaurants by about half—from 19.0 to 9.2 percentage points for Black-owned restaurants and from 8.9 to 5.7 percentage points for Hispanic-owned restaurants. These estimated disparities are still statistically significant. With the full set of controls, there are no statistically significant disparities for Asian- and female-owned restaurants.

Next, we estimate separate regressions for bank and nonbank PPP lenders, with bank results reported in columns 5–8 of Panel A and nonbank results reported in columns 1–4 of Panel B. The findings indicate that the lower rates of PPP borrowing by Black- and Hispanic-owned restaurants are driven by lower rates of PPP borrowing from banks. As shown in column 5 of Panel A, Black-owned restaurants are 34.6 percentage points less likely than white-owned restaurants to receive PPP funding from a bank. Adding controls in column 6, ZIP code fixed effects in column 7, and then restaurant characteristics in column 8 reduces the magnitude of the disparity in half to 16.8 percentage points. Still, the effect remains large and statistically significant. For Hispanic-owned restaurants, the difference is 5.8 percentage points; Asian-owned restaurants — which exhibit no disparity in overall PPP take-up — are 6.1 percentage points less likely to receive a PPP loan from a bank. Female-owned restaurants are 2.6 percentage points less likely to receive a PPP loan from a bank than observably similar male-owned restaurants.

Columns 1–4 of Panel B in Table 3 tell a very different story for PPP loans made by nonbanks. The regressions indicate that for minority-owned restaurants borrowing from nonbanks tends to offset the lower rate of bank PPP borrowing, particularly in the case of restaurants with Black and Asian owners. The regression results reported in column 4, which include ZIP code fixed effects and restaurant characteristics, indicate that Black-owned restaurants are 7.6 percentage points more likely to receive a nonbank PPP loan and Asian-owned restaurants are 6.7 percentage points more likely to receive a nonbank PPP loan. The higher take-up of nonbank PPP loans by Black- and Asian-owned restaurants suggests that these businesses substitute away from banks

Table 3
Racial disparities in the take-up of emergency loans.

Panel A: PPP and Bank PPP

	PPP				Bank PPP			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Black	−0.255***	−0.214***	−0.190***	−0.092***	−0.346***	−0.304***	−0.271***	−0.168***
	(0.023)	(0.024)	(0.026)	(0.025)	(0.022)	(0.023)	(0.025)	(0.024)
Hispanic	−0.107***	−0.090***	−0.089***	−0.057***	−0.113***	−0.096***	−0.092***	−0.058***
	(0.011)	(0.011)	(0.013)	(0.012)	(0.011)	(0.012)	(0.013)	(0.012)
Asian	−0.023*	−0.020	−0.017	0.006	−0.100***	−0.097***	−0.090***	−0.061***
	(0.013)	(0.013)	(0.014)	(0.013)	(0.014)	(0.014)	(0.015)	(0.014)
Other	−0.056**	−0.046**	−0.036	−0.020	−0.094***	−0.084***	−0.069***	−0.050**
	(0.022)	(0.022)	(0.022)	(0.022)	(0.023)	(0.023)	(0.023)	(0.022)
Female	−0.048***	−0.043***	−0.038***	−0.014	−0.059***	−0.054***	−0.051***	−0.026***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.010)	(0.009)
Bank branches per capita		0.024***				0.030***		
		(0.004)				(0.005)		
Ln(Population)		−0.005				−0.002		
		(0.005)				(0.005)		
Ln(Household income)		0.041***				0.040***		
		(0.004)				(0.004)		
COVID cases per capita		−0.002				0.000		
		(0.006)				(0.007)		
Ln(Seats)				0.053***				0.061***
				(0.005)				(0.005)
Ln(Firm age)				0.039***				0.045***
				(0.005)				(0.005)
Accepts credit cards				0.176***				0.145***
				(0.036)				(0.036)
Missing credit cards				−0.003				0.003
				(0.019)				(0.019)
Ln(1 + Reviews)				0.052***				0.053***
				(0.008)				(0.009)
Rating				−0.012				−0.009
				(0.008)				(0.008)
No reviews				−0.100**				−0.090**
				(0.039)				(0.039)
Ln(1 + Page views)				0.008				−0.003
				(0.010)				(0.010)
Ln(1 + Photos)				0.036***				0.048***
				(0.007)				(0.007)
R^2	0.020	0.033	0.109	0.178	0.031	0.044	0.120	0.189
ZIP FEs			✓	✓			✓	✓

Panel B: Nonbank PPP and EIDL

	Nonbank PPP				EIDL			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Black	0.091***	0.090***	0.081***	0.076***	0.025	0.021	0.007	0.022
	(0.016)	(0.016)	(0.017)	(0.017)	(0.023)	(0.023)	(0.025)	(0.026)
Hispanic	0.006	0.006	0.003	0.000	0.066***	0.047***	0.024*	0.023*
	(0.005)	(0.006)	(0.006)	(0.007)	(0.011)	(0.012)	(0.013)	(0.013)
Asian	0.077***	0.077***	0.072***	0.067***	0.024*	0.015	0.008	0.007
	(0.009)	(0.009)	(0.009)	(0.009)	(0.013)	(0.014)	(0.014)	(0.015)
Other	0.038***	0.039***	0.033**	0.030**	0.077***	0.069***	0.061**	0.061**
	(0.013)	(0.013)	(0.013)	(0.013)	(0.023)	(0.023)	(0.024)	(0.024)
Female	0.012**	0.011**	0.013***	0.012**	−0.012	−0.009	−0.001	0.002
	(0.005)	(0.005)	(0.005)	(0.005)	(0.009)	(0.009)	(0.010)	(0.010)
Bank branches per capita		−0.007***				0.020***		
		(0.002)				(0.005)		
Ln(Population)		−0.002				0.025***		
		(0.003)				(0.005)		
Ln(Household income)		0.001				0.005		
		(0.002)				(0.004)		
COVID cases per capita		−0.003				0.030***		
		(0.003)				(0.007)		
Ln(Seats)				−0.008***				0.009*
				(0.003)				(0.005)
Ln(Firm age)				−0.005**				−0.011**
				(0.002)				(0.005)
Accepts credit cards				0.031**				0.088***
				(0.014)				(0.033)
Missing credit cards				−0.005				0.002
				(0.009)				(0.019)

(*continued on next page*)

Table 3 (*continued*).

Ln(1 + Reviews)				−0.002				0.027***
				(0.005)				(0.009)
Rating				−0.003				0.031***
				(0.005)				(0.009)
No reviews				−0.010				0.100**
				(0.021)				(0.040)
Ln(1 + Page views)				0.011**				−0.002
				(0.005)				(0.011)
Ln(1 + Photos)				−0.011***				−0.010
				(0.004)				(0.007)
R^2	0.015	0.016	0.089	0.092	0.003	0.008	0.079	0.083
ZIP FEs			✓	✓			✓	✓

This table reports the results of linear probability model regressions of receiving emergency loans:

$$Emergency\ loan_{f,r,z} = \alpha_z + \beta \cdot Minority_f + \delta \cdot Female_f + \gamma' X_{f,r} + \varepsilon_{f,r,z},$$

where f indexes firms, r indexes restaurants and z indexes ZIP codes. The sample construction is summarized in Tables A.2 and 1. Robust standard errors are reported. *, **, and *** indicate statistical significance at 10%, 5%, and 1%. Continuous variables are standardized so that their coefficients represent the effect of a one standard deviation change. $N = 12,812$, except for columns 2 and 6 where $N = 12,764$ due to missing ZIP-level controls.

because they find it more difficult or less desirable to access PPP loans from banks. Note that the substitution to nonbank PPP loans is large enough for Asian-owned restaurants to fully offset their lower rate of borrowing from banks, but this is not the case for Black-owned restaurants, which have lower rates of overall PPP take-up. The restaurant controls also indicate that smaller and younger restaurants substitute away from banks, which tend to lend more to larger and older firms.

In columns 5–8 of Panel B, we present the results on the determinants of EIDL borrowing. As noted above, firms applied for EIDL loans directly through the SBA website. They could apply for and receive loans from both the PPP and EIDL program at the same time, though the proceeds had to be used for different purposes. In the column 8 regression, which includes controls for location and restaurant characteristics, Black-owned restaurants are 2.2 percentage points more likely than white-owned restaurants to receive EIDL loans, though the difference is not statistically significant. Hispanic-owned restaurants are 2.3 percentage points more likely than white-owned restaurants to receive loans from EIDL, which is statistically significant at 10%.

The fact that minority-owned restaurants are more likely to receive emergency loans from nonbanks and, to some extent, from the EIDL program suggests that lower demand for emergency loans by minority-owned restaurants does not explain racial disparities in bank PPP lending. In Section 4, we more fully examine the possibility that our findings are explained by minority-owned restaurants having lower demand for emergency loans. For example, minority-owned restaurants may have had lower demand for PPP loans because they were more likely to shut down during the pandemic or were operating at reduced capacity relative to white-owned restaurants. This might be the case if minority-owned restaurants were less profitable going into the pandemic or were more likely to serve minority populations more adversely affected by the pandemic. It is also possible that minority-owned restaurants were less aware of emergency support programs or were less willing to apply to such programs. We present a variety of results that allow us to reject these alternative explanations.

Before we move on to analyze potential differences in demand, we note that our findings are robust to a number of alternative sample specifications. In columns 1 and 4 of the Internet Appendix Table IA1, we restrict the sample to single-restaurant firms, which comprise 93% of the sample.[20] We find very similar results for PPP loans (column 1) and bank PPP loans (column 4). In columns 2 and 5, we redefine the dependent variable to be equal to one if either the firm or any of its affiliates receives a bank PPP loan. We consider two firms to be affiliated if they have the same first officer.[21] We find somewhat stronger results when incorporating information about PPP loans received by affiliates. Finally, in columns 3 and 6 we examine robustness to using fixed effects for Census block groups, a much more granular geography than a ZIP code. This regression is meant to address the concern that minority-owned restaurants may have been located farther away from bank branches or may have been subject to different shocks even within the same ZIP code.[22] We find similar results when comparing minority- and white-owned restaurants within these more narrowly defined geographic areas.

4. Demand for PPP loans

As previously noted in Section 3 above, an alternative explanation of our finding of disparities in PPP take-up is that, despite our best efforts, we have not adequately controlled for differences in firm characteristics that could affect the demand for PPP loans. This alternative explanation is inconsistent with the finding that minority-owned businesses make greater use of nonbank PPP loans and EIDL loans. Nevertheless, to further and more directly address any remaining concerns, this section explores whether minority-owned restaurants had lower demand for emergency loans. We first examine whether observable differences between minority- and white-owned restaurants during the pandemic could have driven a difference in the demand for emergency loans. We then restrict the sample to restaurants with demonstrated demand for emergency support to see whether the estimated disparities exist within this subsample. Both approaches suggest that differential demand cannot explain our findings.

4.1. Observable differences in demand for PPP loans

In this section, we examine the hypothesis that minority-owned restaurants had lower demand for PPP loans by exploring the most plausible mechanisms that could have driven lower demand. In particular, minority-owned restaurants may have been more likely to shut down during the pandemic or operate at reduced capacity relative to white-owned restaurants. This might be the case if minority-owned restaurants were less profitable going into the pandemic or were more likely to serve minority populations more adversely affected by the pandemic.

Table 4 tries to address this concern in a number of ways. In this table, we include as additional controls several measures of restaurant

[20] This classification is based on the restaurant's license holder (immediate owner) and does not account for restaurants owned by affiliated firms.

[21] Specifically we match on the first officer's first and last names, city, ZIP code, and street address.

[22] We obtain 2010 Census block information from the Census geocoder and Texas A&M GeoServices. Because the Census geocoder has weak coverage of non-residential addresses, we first run restaurant addresses through the Census geocoder, and then run the unmatched addresses through Texas A&M GeoServices. There are 832 ZIP codes and 4,740 Census block groups in the data.

Table 4
Controlling for activity during the pandemic.

Panel A: PPP and Bank PPP								
	PPP				Bank PPP			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Black	−0.103***	−0.092***	−0.103***	−0.119***	−0.175***	−0.165***	−0.172***	−0.176***
	(0.026)	(0.027)	(0.027)	(0.030)	(0.025)	(0.025)	(0.026)	(0.029)
Hispanic	−0.055***	−0.057***	−0.054***	−0.048***	−0.053***	−0.055***	−0.049***	−0.046***
	(0.012)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.014)
Asian	0.005	0.004	0.002	0.004	−0.061***	−0.062***	−0.063***	−0.066***
	(0.013)	(0.014)	(0.014)	(0.015)	(0.014)	(0.015)	(0.015)	(0.016)
Other	−0.020	−0.012	−0.012	−0.024	−0.049**	−0.034	−0.033	−0.051**
	(0.022)	(0.022)	(0.022)	(0.023)	(0.023)	(0.023)	(0.023)	(0.024)
Female	−0.013	−0.017*	−0.016*	−0.016	−0.025***	−0.031***	−0.030***	−0.032***
	(0.009)	(0.009)	(0.009)	(0.010)	(0.009)	(0.010)	(0.010)	(0.010)
Ln(Reviews during pandemic)	0.013*		0.009	0.008	0.026***		0.021***	0.018**
	(0.007)		(0.007)	(0.008)	(0.007)		(0.008)	(0.008)
Ln(Visits during pandemic)		0.054***	0.018	0.004		0.049***	0.018	0.007
		(0.013)	(0.014)	(0.016)		(0.013)	(0.015)	(0.017)
Ln(Visits before pandemic)		−0.040***	−0.008	0.007		−0.034***	−0.006	0.005
		(0.013)	(0.014)	(0.016)		(0.013)	(0.014)	(0.016)
N	12,464	11,909	11,602	10,517	12,464	11,909	11,602	10,517
R^2	0.174	0.182	0.179	0.174	0.185	0.194	0.190	0.185
Panel B: Nonbank PPP and EIDL								
	Nonbank PPP				EIDL			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Black	0.071***	0.073***	0.069***	0.057***	0.016	0.031	0.020	0.034
	(0.017)	(0.018)	(0.018)	(0.019)	(0.027)	(0.027)	(0.028)	(0.030)
Hispanic	−0.002	−0.002	−0.005	−0.002	0.025*	0.021	0.022	0.019
	(0.007)	(0.007)	(0.007)	(0.007)	(0.014)	(0.014)	(0.014)	(0.015)
Asian	0.066***	0.066***	0.065***	0.069***	0.003	0.003	−0.001	−0.002
	(0.009)	(0.010)	(0.010)	(0.010)	(0.015)	(0.015)	(0.015)	(0.016)
Other	0.029**	0.022	0.020	0.027*	0.061**	0.070***	0.069***	0.054**
	(0.013)	(0.013)	(0.013)	(0.014)	(0.024)	(0.025)	(0.025)	(0.026)
Female	0.012**	0.014***	0.014**	0.016***	−0.001	−0.006	−0.009	−0.014
	(0.005)	(0.005)	(0.005)	(0.006)	(0.010)	(0.010)	(0.010)	(0.011)
Ln(Reviews during pandemic)	−0.013***		−0.012***	−0.011**	−0.023***		−0.022***	−0.028***
	(0.004)		(0.004)	(0.004)	(0.007)		(0.008)	(0.008)
Ln(Visits during pandemic)		0.005	0.000	−0.003		0.001	−0.014	−0.026
		(0.006)	(0.007)	(0.007)		(0.014)	(0.015)	(0.017)
Ln(Visits before pandemic)		−0.006	−0.001	0.002		−0.003	0.010	0.021
		(0.006)	(0.006)	(0.007)		(0.014)	(0.015)	(0.017)
N	12,464	11,909	11,602	10,517	12,464	11,909	11,602	10,517
R^2	0.094	0.096	0.098	0.105	0.087	0.088	0.092	0.102
ZIP FEs	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Exclude closed restaurants				✓				✓

This table shows the robustness of the results to controlling for restaurant activity during the pandemic and to excluding restaurants that closed permanently. The sample construction is summarized in Tables A.2 and 1. Columns 4 and 8 exclude restaurants that closed during the pandemic. Ln(Reviews during pandemic) is the natural log of one plus the number of Yelp reviews posted between March 11 and August 8, 2020. Ln(Visits during pandemic) is the natural log of one plus the average number of monthly visits during March–July 2020. Ln(Visits before pandemic) is the natural log of one plus the average number of monthly visits during October 2019–February 2020. Monthly visits are from SafeGraph. Controls include all restaurant characteristics included in the regression in column 4 of Table 3. Robust standard errors are reported. *, **, and *** indicate statistical significance at 10%, 5%, and 1%. Continuous variables are standardized so that their coefficients represent the effect of a one standard deviation change.

activity during the pandemic to see whether they could help explain racial disparities. Note that while activity levels may explain PPP borrowing, it is also possible that PPP borrowing could explain activity levels, as a PPP loan could help a restaurant stay in business. Thus, including pandemic activity levels could lead us to underestimate racial disparities in PPP borrowing.

Our first measure of restaurant activity is the natural log of one plus the number of Yelp reviews posted during the pandemic, specifically between March 11, 2020, when the World Health Organization declared COVID a pandemic, and August 8, 2020, when the second round of the PPP closed. Restaurants with more reviews are more likely to receive PPP loans and bank PPP loans but are less likely to receive nonbank PPP loans and EIDL loans, as seen in columns 1 and 5 in Panels A and B of Table 4. These regressions also include the full set of controls shown in column 4 of Table 3. Recall that these controls include measures of pre-pandemic activity, such as the number of pre-pandemic restaurant reviews. Notably, while this finding indicates that more active restaurants are more likely to receive bank PPP funding, this control has no meaningful effect on the estimated coefficients of the minority-owned restaurant variables.

Our second measure of restaurant activity during the pandemic is the natural log of one plus the number of average monthly restaurant visits. These data come from SafeGraph, which collects anonymous data on visits to points of interest using cell phone tracking.[23] We also include the average number of monthly visits to the restaurant

[23] SafeGraph partners with mobile apps that obtain consent from their users to collect location data. SafeGraph uses the raw geographic coordinates of a user's cell phone to determine the store, restaurant, or another point of interest that the user was visiting when their cell phone pinged.

before the pandemic as another control for pre-pandemic profitability. Columns 2 and 6 in both panels confirm the finding that restaurant activity during the pandemic is associated with a higher likelihood of receiving PPP funding from banks, but it does not affect the estimated coefficients of the minority-owned restaurant variables. Columns 3 and 7 use both measures of restaurant activity during the pandemic, namely Yelp reviews and restaurant visits. In column 3, neither measure is statistically significant, likely because of multicollinearity, while in column 7 only Yelp reviews are statistically significant. Importantly, estimated racial disparities are not materially affected.

Finally, in columns 4 and 8 of both panels, we again include the above measures of restaurant activity during the pandemic, but we exclude restaurants that closed after the pandemic began. We consider a restaurant to have permanently closed if at least one of the following three conditions is satisfied: (1) Yelp reports a valid permanent closure date; (2) restaurant license data indicate a change in ownership taking place after February 15, 2020; (3) the restaurant is not listed as active in the October 2021 restaurant licenses data. According to these criteria, about 11% of restaurants in our data have permanently closed or changed ownership since the pandemic. Excluding closed restaurants from our analysis does not materially affect the estimated racial disparities.

4.2. Unobservable differences in demand for PPP loans

Although our regression specifications control for an extensive set of observable restaurant characteristics — both before and during the pandemic — there are possibly unobservable differences in demand that could help explain racial disparities in PPP borrowing. One possibility is that minority-owned restaurants were less aware of the PPP. Indeed, in a survey conducted by the Federal Reserve, 20% of Black-owned businesses cited a lack of awareness of the PPP as a reason why they did not apply; only 7% of white-owned non-applicants cited this reason (Federal Reserve Banks, 2021). Even after controlling for observable differences between Black- and white-owned firms, Black-owned firms are 5% more likely to state this reason for not applying (Chernenko et al., 2024). Another possibility is that minority-owned restaurants may have had fewer employees even controlling for location and restaurant activity. Since the size of a PPP loan is based on payroll, restaurants with fewer employees receive smaller loans and thus may have found that the costs of applying for a PPP loan outweighed the benefits.

We address these concerns by exploiting data on the EIDL Advance program.[24] The EIDL Advance program provided grants of $1,000 per employee, up to 10 employees. About half of the restaurants in our sample received EIDL Advance grants. These restaurants were obviously aware of the EIDL program and thus very likely to have been aware of other emergency relief programs like the PPP. At the same time, at $1,000 per employee, EIDL Advance grants were very modest compared to the PPP, which provided forgivable loans of 2.5 times monthly employee wages. PPP loans would exceed the size of the EIDL Advance grant provided a worker earned more than $400 per month, a very modest amount. Although applying for a PPP loan after receiving an EIDL Advance grant involved additional costs and although PPP loans came with some strings attached to them, 80% of restaurants in our sample that received an EIDL Advance grant also applied for and received a PPP loan.[25] Thus, by examining racial disparities in PPP borrowing among restaurants that received EIDL Advance grants, our analysis focuses on a sample of firms that wanted emergency loans and were aware of their availability.

Restricting the sample to EIDL Advance recipients has the added benefit of allowing us to control for the number of employees, which should be closely correlated with payroll. Because the grant size is $1,000 times the number of employees up to $10,000, we can back out the number of employees up to 10. Because EIDL Advance grants are capped at $10,000, we include an indicator variable for firms with at least ten employees. As an alternative approach, we restrict the sample to firms with fewer than 10 employees.

Table 5 reports the results of linear probability model regressions of receiving PPP loans (columns 1–2), bank PPP loans (columns 3–4), and nonbank PPP loans (columns 5–6), conditional on receiving an EIDL Advance grant. Column 1 of Table 5 repeats the baseline specification in column 4 of Table 3, but also includes the number of employees implied by the EIDL Advance grant up to 10 and an indicator variable for whether the EIDL Advance grant was the maximum amount of $10,000. An additional employee is associated with a 5.1 percentage point greater probability of receiving a PPP loan, but beyond 10 or more employees there is no incremental increase in the likelihood of receiving a PPP loan. Notably, the disparities in PPP take-up remain large: Black-owned restaurants are 11.8 percentage points less likely to receive PPP loans controlling for the number of employees.

Columns 1–2 show that PPP take-up is increasing in the number of employees. This effect is driven by a positive relationship between bank PPP loans and the number of employees (columns 3–4). This is not the case for nonbank PPP loans (columns 5–6), suggesting that the relationship between PPP take-up and the number of employees is likely driven by the supply of bank PPP loans rather than by the demand for PPP funding. The even-numbered columns further restrict the sample to firms with fewer than 10 employees. The results in these columns are similar to those in the odd-numbered columns.[26]

Finding racial disparities in PPP borrowing in the subsample of EIDL Advance participants shows that racial disparities in the PPP cannot be explained by lower demand for emergency support by minority-owned businesses or less awareness of such support programs. Another way to establish this is to look at the subsample of restaurants that receive EIDL or PPP loans. Similar to our analysis of EIDL Advance program participants, we can ask whether there are racial disparities in PPP take-up among the sample of firms that receive EIDL or PPP loans. These restaurants have a demonstrated demand for emergency loans and an awareness that such programs exist. In Table 6, we find that there are racial disparities in PPP take-up within this subsample of firms, with Black- and Hispanic-owned restaurants both significantly less likely to receive PPP loans. As before, the disparities are driven by disparities in bank borrowing. Minority-owned restaurants are less likely to borrow from banks even controlling for ZIP code and restaurant characteristics. In particular, Black-owned restaurants receiving an emergency loan are 20.5 percentage points less likely than white-owned restaurants to receive a PPP loan from a bank. This finding further supports the view that unobserved differences across groups in emergency loan demand or loan awareness do not explain disparities in PPP borrowing.

5. Economic mechanisms

Having ruled out differences in demand as primary drivers of conditional disparities in the take-up of PPP loans, we turn in this section to examining three economic mechanisms: bank relationships, access to professional services, and racial bias.

[24] https://data.sba.gov/dataset/covid-19-eidl-advance.

[25] An EIDL Advance grant would count against the forgivable portion of the PPP loan.

[26] In untabulated results, we combine the approaches in Tables 4 and 5 to control for demand. In other words, we control for the Yelp and SafeGraph measures of activity during the pandemic in the regressions looking at PPP take-up among restaurants receiving EIDL Advance grants. We find similar disparities. However, measures of activity during the pandemic are not statistically significant within this sample.

Table 5
PPP borrowing conditional on receiving EIDL advance.

	PPP		Bank PPP		Nonbank PPP	
	(1)	(2)	(3)	(4)	(5)	(6)
Black	−0.118***	−0.089*	−0.215***	−0.193***	0.097***	0.104***
	(0.038)	(0.053)	(0.037)	(0.049)	(0.028)	(0.039)
Hispanic	−0.050***	−0.038	−0.037**	−0.029	−0.014	−0.009
	(0.015)	(0.028)	(0.017)	(0.030)	(0.010)	(0.017)
Asian	0.026	0.046*	−0.050***	−0.044	0.075***	0.089***
	(0.016)	(0.028)	(0.019)	(0.030)	(0.014)	(0.021)
Other	−0.020	−0.046	−0.061**	−0.080	0.041**	0.034
	(0.027)	(0.048)	(0.029)	(0.049)	(0.020)	(0.032)
Female	−0.015	−0.017	−0.032**	−0.044**	0.017**	0.027*
	(0.012)	(0.021)	(0.013)	(0.022)	(0.008)	(0.014)
Num. employees	0.051***	0.046***	0.049***	0.045***	0.003	0.001
	(0.004)	(0.005)	(0.004)	(0.005)	(0.003)	(0.003)
I(Num. employees ≥ 10)	−0.074***		−0.050**		−0.024	
	(0.022)		(0.024)		(0.015)	
N	6277	2831	6277	2831	6277	2831
R^2	0.292	0.335	0.296	0.325	0.157	0.273
ZIP FEs	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓
Exclude firms with ≥ 10 employees		✓		✓		✓

In this table we restrict the sample to restaurants that received EIDL Advance grants and use grant size as a proxy for the number of employees. The EIDL Advance program provided grants of $1,000 per employee up to a maximum of $10,000. Odd-numbered columns include a dummy for firms with 10 or more employees, i.e., firms receiving EIDL Advance grants of $10,000. Even-numbered columns exclude firms receiving EIDL Advance grants of $10,000. The sample construction is summarized in Tables A.2 and 1. Controls include all restaurant characteristics included in the regression in column 4 of Table 3. Robust standard errors are reported. *, **, and *** indicate statistical significance at 10%, 5%, and 1%.

Table 6
Demand for credit.

	PPP			Bank PPP		
	(1)	(2)	(3)	(4)	(5)	(6)
Black	−0.157***	−0.115***	−0.068**	−0.323***	−0.270***	−0.205***
	(0.025)	(0.027)	(0.027)	(0.029)	(0.032)	(0.031)
Hispanic	−0.075***	−0.062***	−0.051***	−0.088***	−0.071***	−0.054***
	(0.009)	(0.010)	(0.009)	(0.010)	(0.012)	(0.012)
Asian	−0.007	−0.001	0.012	−0.109***	−0.096***	−0.073***
	(0.008)	(0.009)	(0.009)	(0.013)	(0.014)	(0.014)
Other	−0.051***	−0.039**	−0.031*	−0.102***	−0.081***	−0.069***
	(0.017)	(0.017)	(0.017)	(0.022)	(0.022)	(0.022)
Female	−0.027***	−0.027***	−0.014**	−0.045***	−0.046***	−0.028***
	(0.007)	(0.007)	(0.007)	(0.009)	(0.009)	(0.009)
R^2	0.021	0.116	0.154	0.038	0.134	0.176
ZIP FEs		✓	✓		✓	✓
Controls			✓			✓

This table reports the results of linear probability model regressions of receiving a PPP loan on the restaurant owner's race, limiting the sample to firms that receive any type of emergency loans:

$$PPP_{f,r,z} = \alpha_z + \beta \cdot Minority_f + \delta \cdot Female_f + \gamma' X_{f,r} + \varepsilon_{f,r,z},$$

where f indexes firms, r indexes restaurants, and z indexes ZIP codes. The sample consists of restaurants that receive either PPP or EIDL loans. Controls include all restaurant characteristics included in the regression in column 4 of Table 3. Robust standard errors are reported. *, **, and *** indicate statistical significance at 10%, 5%, and 1%. $N = 9,784$.

5.1. Bank relationships

This section explores the role that bank relationships could play in explaining racial disparities in PPP borrowing. Banks may have been more likely to help their existing clients complete their PPP applications or correct mistakes in the submitted applications. If minority-owned businesses had relatively weak relationships with banks, then we would expect to see racial disparities in bank PPP borrowing without adequate controls for these relationships. Numerous studies have shown that Black- and Hispanic-owned businesses are more likely to be turned down for loans, controlling for credit quality and personal wealth, and less likely to apply for loans (Blanchflower et al., 2003; Cavalluzzo and Wolken, 2005; Fairlie et al., 2022). In addition, minority-owned businesses report greater dissatisfaction with their financial services providers (Federal Reserve Banks, 2021). This evidence suggests that, relative to white-owned restaurants, minority-owned restaurants may have weaker relationships with banks.

While we do not observe all aspects of a restaurant's banking relationships, restaurants in the sample likely have some kind of banking relationship because they must have a bank account to receive credit card payments. Nevertheless, some may have stronger bank relationships than others because they use a wider variety of banking services or they have been bank customers for a longer time. To get at the strength of a firm's relationship with a bank, we collect data on whether a firm has outstanding bank loans. Although we cannot observe all of a firm's loans, we can observe secured loans except those secured by real estate. According to Luck and Santos (2023), out of all loans to firms with less than $50 million in assets by banks that are subject to the Federal Reserve's stress tests, only 3.6% are unsecured and 22% are secured by real estate. Furthermore, Blanchflower et al. (2003) show that Black-owned firms borrow with credit cards at the same rate as white-owned firms. Thus, it is likely that by observing non-mortgage secured loans, we are observing the most meaningful cross-sectional variation in a firm's borrowing.

As noted in the data section, we use Florida Uniform Commercial Code (UCC) financing statements to measure secured borrowing relationships between restaurants and their lenders, both banks and nonbanks. Lenders file these financing statements to assert their security interest in a loan. The security interest could be in physical capital, such as equipment, or a general lien on the business. We consider a restaurant to have a secured borrowing relationship with a lender if there was an active UCC filing by that lender as of February 15, 2020. As shown in Table 2, the percentage of Black-owned restaurants that have UCC loans with banks (4%) is considerably lower than it is for white-owned restaurants (19%), although the percentages are quite similar for loans from nonbanks (14%).

The dependent variable in columns 1–2 of Table 7 is an indicator for whether or not the restaurant receives a PPP loan. Columns 3–4 report the results for bank PPP loans, while columns 5–6 report the results for nonbank PPP loans. In columns 7–8 we present the regression results for EIDL loans. Odd-numbered columns repeat the baseline specifications from columns 4 and 8 of both panels in Table 3. Even-numbered columns add controls for whether the firm had an

Table 7
UCC borrowing relationships.

	PPP		Bank PPP		Nonbank PPP		EIDL	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Black	−0.092***	−0.091***	−0.168***	−0.165***	0.076***	0.074***	0.022	0.020
	(0.025)	(0.026)	(0.024)	(0.024)	(0.017)	(0.017)	(0.026)	(0.026)
Hispanic	−0.057***	−0.056***	−0.058***	−0.055***	0.000	−0.001	0.023*	0.025*
	(0.012)	(0.012)	(0.012)	(0.012)	(0.007)	(0.006)	(0.013)	(0.013)
Asian	0.006	0.012	−0.061***	−0.055***	0.067***	0.067***	0.007	0.017
	(0.013)	(0.013)	(0.014)	(0.014)	(0.009)	(0.009)	(0.015)	(0.015)
Other	−0.020	−0.018	−0.050**	−0.049**	0.030**	0.031**	0.061**	0.066***
	(0.022)	(0.022)	(0.022)	(0.022)	(0.013)	(0.013)	(0.024)	(0.024)
Female	−0.014	−0.013	−0.026***	−0.024***	0.012**	0.011**	0.002	0.004
	(0.009)	(0.009)	(0.009)	(0.009)	(0.005)	(0.005)	(0.010)	(0.010)
Bank UCC loan		0.055***		0.083***		−0.028***		0.057***
		(0.010)		(0.011)		(0.005)		(0.013)
Nonbank UCC loan		0.042***		0.009		0.033***		0.104***
		(0.011)		(0.012)		(0.007)		(0.013)
R^2	0.178	0.180	0.189	0.192	0.092	0.095	0.083	0.091
ZIP FEs	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓

This table reports the results of linear probability model regressions of receiving different types of emergency loans on the restaurant owner's race and controls for existing UCC borrowing relationships. The sample construction is summarized in Tables A.2 and 1. Controls include all restaurant characteristics included in the regression in column 4 of Table 3. Robust standard errors are reported. *, **, and *** indicate statistical significance at 10%, 5%, and 1%. $N = 12,812$.

active bank or nonbank UCC loan. All regressions include ZIP code fixed effects and the full set of restaurant characteristics from Table 3, although they are not shown.

The results reported in column 2 indicate that having a bank UCC loan is associated with a 5.5 percentage point higher probability of receiving a PPP loan. Having a nonbank UCC loan increases the probability of receiving a PPP loan by 4.2 percentage points, which is also statistically significant. These effects are fairly large; given that only 30% of restaurants do not receive PPP loans, a 5.5 percentage points increase in the probability of receiving a PPP loan reduces the likelihood of not receiving a PPP loan by about 18%.

Controlling for secured borrowing relationships does not affect the coefficient estimates of the minority restaurant-owner dummies. Black-owned restaurants are still 9.1 percentage points less likely than white-owned restaurants to receive a PPP loan, even after controlling for existing UCC loans.

Given these findings, it is not surprising that UCC borrowing relationships increase the likelihood of receiving a PPP loan from a bank. As shown in column 4, restaurants are 8.3 percentage points more likely to receive a bank PPP loan if they have an outstanding bank UCC loan. Having a nonbank UCC loan has no effect on the probability of receiving a bank PPP loan. As in the overall PPP regressions, the inclusion of the UCC loan controls has almost no effect on the Black-owned restaurant dummy and has only a very modest effect on the coefficients of the other minority-owned restaurant dummies, with the strongest effect being the reduction in the coefficient on *Asian* from −6.1 to −5.5 percentage points.

In the regressions of nonbank PPP loans in columns 5–6, we find that restaurants that have bank UCC loans are 2.8 percentage points less likely to receive PPP loans from nonbanks. This result follows from our finding that restaurants with bank UCC loans are more likely to receive PPP loans from banks. Restaurants that have nonbank UCC loans are 3.3 percentage points more likely to receive PPP loans from nonbanks.

Finally, in columns 7–8 we find that restaurants with bank and nonbank UCC loans are 5.7 and 10.4 percentage points more likely to receive EIDL loans. This result likely reflects refinancing of existing bank and nonbank loans using EIDL loans.

Importantly, controlling for UCC borrowing relationships has little effect on the estimated racial disparities in the take-up of PPP loans. Thus, although minority-owned restaurants are less likely to have bank UCC loans that are associated with a higher probability of receiving PPP loans from banks, this effect is not large enough to explain disparities in the take-up of PPP loans.

Existing UCC borrowing relationships could have different effects for minority- and white-owned firms. If minority-owned firms without existing bank relationships have a harder time getting help from banks than white-owned firms without such relationships, then existing bank relationships may have a stronger effect on the probability that minority-owned firms receive PPP loans from banks. On the other hand, among their existing clients, banks may prioritize white-owned firms. Minority-owned businesses, for example, report greater dissatisfaction with their financial services providers (Federal Reserve Banks, 2021). Unfortunately, our restaurants sample has relatively few minority-owned firms with existing bank relationships. There are only 20 Black-owned restaurants with existing bank relationships. The small number of minority-owned firms with existing bank relationships makes it difficult to reliably estimate differential effects of UCC borrowing relationships. We therefore postpone the examination of differential effects of UCC borrowing relationships until Section 6, where we study the take-up of PPP loans in a much larger sample of firms across all industries.

5.1.1. *Home mortgages*

Another dimension of bank relationships is home mortgages taken out by the firm's owners. To measure such relationships, we search county records for mortgages taken out by the firm's owners. This search is limited to officers that were previously matched to registered voters. Because we match officers and mortgagors exclusively on name, we exclude cases where multiple registered voters in a county share the same name. As a result, we cannot measure home mortgage borrowing relationships for some firms in our sample. Overall, we are able to measure mortgage borrowing for 9,967 out of 12,812 restaurant owners in the sample.

Because it is difficult to track mortgages over time, we define our bank (nonbank) mortgage variable to be an indicator for whether any of the firm's officers took out a mortgage from a bank (nonbank) lender over the ten years before February 15, 2020. Thus it is possible that some owners had repaid their mortgages before February 15, 2020 and had no active relationship at the time. It is also possible that some owners had active mortgages that they took out before February 15, 2010, and that they had not refinanced since then. Finally, note that we limit our search to mortgages in the county in which an officer is registered as a voter. It is possible that some officers have homes and mortgages in other counties.

In our sample, 41% of restaurants have at least one officer who has taken out a home mortgage during the ten years before the pandemic.

Table 8
Controlling for home mortgages.

	PPP		Bank PPP		Nonbank PPP		EIDL	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Black	−0.075**	−0.072**	−0.145***	−0.139***	0.070***	0.068***	0.022	0.020
	(0.029)	(0.029)	(0.028)	(0.028)	(0.019)	(0.019)	(0.029)	(0.029)
Hispanic	−0.051***	−0.050***	−0.047***	−0.046***	−0.003	−0.004	0.039**	0.037**
	(0.014)	(0.014)	(0.015)	(0.015)	(0.007)	(0.007)	(0.016)	(0.016)
Asian	0.006	0.008	−0.060***	−0.057***	0.066***	0.065***	0.015	0.015
	(0.016)	(0.016)	(0.017)	(0.017)	(0.010)	(0.011)	(0.017)	(0.017)
Other	−0.010	−0.009	−0.047*	−0.046*	0.038***	0.037***	0.037	0.038
	(0.023)	(0.023)	(0.024)	(0.024)	(0.014)	(0.014)	(0.026)	(0.026)
Female	−0.014	−0.013	−0.021**	−0.018*	0.007	0.005	0.004	0.003
	(0.010)	(0.010)	(0.010)	(0.010)	(0.006)	(0.006)	(0.011)	(0.011)
Bank UCC loan	0.056***	0.052***	0.087***	0.081***	−0.031***	−0.028***	0.065***	0.066***
	(0.012)	(0.012)	(0.012)	(0.012)	(0.006)	(0.006)	(0.015)	(0.015)
Nonbank UCC loan	0.053***	0.055***	0.019	0.021	0.035***	0.034***	0.114***	0.113***
	(0.012)	(0.012)	(0.013)	(0.013)	(0.008)	(0.008)	(0.015)	(0.015)
Bank mortgage		0.020**		0.039***		−0.019***		−0.015
		(0.010)		(0.011)		(0.005)		(0.012)
Nonbank mortgage		0.027**		0.035***		−0.008		0.022
		(0.012)		(0.012)		(0.006)		(0.014)
R^2	0.194	0.195	0.205	0.207	0.121	0.122	0.109	0.110
ZIP FEs	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓

This table reports the results of linear probability model regressions of receiving different types of emergency loans on the restaurant owner's race and controls for firm owners having bank and nonbank home mortgages. For each officer or director, we search county records for mortgages entered into during the ten year period ending February 15, 2020. We classify lenders into bank versus nonbank based on the lender name. The number of observations (9,967) is smaller than in the full sample (12,812) because (a) when there are multiple voters in a county with the same first and last names, we cannot unambiguously match firm officers with mortgage borrowers and (b) not all county systems report lender names, preventing us from classifying lenders as banks versus nonbanks. All regressions are estimated on the same sample. Controls include all restaurant characteristics included in the regression in column 4 of Table 3. Robust standard errors are reported. *, **, and *** indicate statistical significance at 10%, 5%, and 1%. $N = 9,967$.

32% of restaurants had taken out a mortgage from a bank, while 18% had taken out a mortgage from a nonbank.[27]

Table 8 reports the results. In columns 1–2, the dependent variable is an indicator for whether or not the restaurant receives a PPP loan. Column 1 re-estimates the regression specification in column 2 of Table 7 but limits the sample to restaurants for which we can measure mortgage borrowing relationships. Column 2 adds controls for bank and nonbank mortgages. Controlling for observable firm characteristics, firms whose owners took out home mortgages from bank (nonbank) lenders during the previous ten years are 2.0 (2.7 percenage points) more likely to receive PPP loans. Crucially, controlling for home mortgages has a fairly small effect on the estimated coefficient on *Black*, which declines in magnitude from −7.5 to −7.2 percentage points.

In columns 3–4, the dependent variable is an indicator for whether or not the restaurant receives a bank PPP loan. Controlling for observable firm characteristics, firms whose owners took out home mortgages from bank (nonbank) lenders during the previous ten years are 3.9 (3.5) percentage points more likely to receive bank PPP loans. Controlling for home mortgages once again has a small effect on the estimated coefficient on *Black*, which declines in magnitude from −14.5 to −13.9 percentage points.

Columns 5–6 report the results for the take-up of nonbank PPP loans, while columns 7–8 report the results for the take-up of EIDL loans. According to column 6, taking out a bank mortgage is associated with a 1.9 percentage point reduction in the probability of getting a nonbank PPP loan, presumably because these restaurants are more likely to get bank PPP loans. In contrast to UCC loans, mortgage borrowing relationships have no effect on the take-up of EIDL loans. This makes sense since firms could use EIDL to refinance existing business loans but not personal loans and since mortgage borrowing relationships are unlikely to affect approval of EIDL loans.

Overall, the results in Tables 7 and 8 show that while existing borrowing relationships are associated with greater take-up of PPP loans, the estimated effects are too small to explain racial disparities in PPP take-up. One might be concerned, however, that we measure with error whether a restaurant owner has a mortgage and such measurement error could bias the estimated coefficients on bank and nonbank mortgages towards zero. This would, in turn, cause us to underestimate the extent to which more precisely measured home mortgage controls would reduce estimated racial disparities in the take-up of PPP loans. Although this concern may be valid in theory, our back-of-the-envelope calculations described below suggest that the effect is small and is unlikely to result in an economically significant reduction in the estimated disparity in Table 8.

To see this, note that the extent to which the coefficient on *Black* declines in magnitude once we control for home mortgages is determined by the product of (a) the coefficient on bank (nonbank) mortgages and (b) the difference in the likelihood that Black and white restaurant owners have a bank (nonbank) mortgage, conditional on the observable characteristics included in the regression. For example, in the regression in column 2 of Table 8, the estimated coefficient on the bank mortgage indicator is 0.020. Given a conditional disparity of 0.14 in bank mortgages, the reduction in the magnitude of the coefficient on *Black* is $0.14 \times 0.020 = 0.0028$. To evaluate the effect of measurement error in the home mortgage variables, note that our algorithm for determining which firms had banking relationships through home mortgages taken out by their owners is unlikely to generate false positives; if we identify an owner as having a home mortgage, it is very likely to be the case. However, we may fail to tag a firm's owner as having a mortgage when he or she actually has one. Thus, suppose that the true rate of owners with bank home mortgages is 64%, or double our estimated prevalence in the sample. This is likely to be a very conservative upper bound given that the home-ownership rate is around 65% and some homeowners either do not have a mortgage or have one from a nonbank. Furthermore, suppose that the true (conditional) disparity in bank home mortgages is 28%, double the estimated 14% disparity in our data. On the assumption that we are underestimating by 50% the share of owners with bank home mortgages, the true effect of bank home mortgages would likely be about twice as large as the

[27] A firm's officers could have taken out mortgages from both bank and nonbank lenders.

estimated coefficient.[28] Under this very conservative set of assumptions, the true effect of controlling for bank mortgages in the PPP regressions in column 2 of Table 8 is around $0.28 \times (2 \times 0.020) = 0.0112$. This is almost an order of magnitude smaller than the estimated disparity of 0.075.[29] Thus, measurement error in home mortgages is unlikely to have a significant effect on our results.

5.2. Access to professional services

While the PPP application itself was short, there were a number of documentation requirements that many small businesses had difficulty complying with. For example, as described in Chernenko et al. (2024), sole proprietors with paid employees had to submit Form 1040, Schedule C, along with four quarters of employer's quarterly federal tax return (Form 941), four quarters of state unemployment tax returns, and proof of contributions to any benefits programs. Some lenders also required other documents to comply with the Bank Secrecy Act and Anti-Money Laundering regulations. In addition, there were some ambiguities — particularly early on in the program — about what qualified as eligible expenses. Some of the instructions for determining the eligible amount could be difficult to interpret for less financially-sophisticated small business owners. Thus, many businesses relied on accountants and lawyers, as well as other professionals, to help them fill out PPP applications and submit the required documents. Firms that previously used accountants and lawyers were also likely better positioned to access the documentation required for the application.

In this section, we ask whether the lower take-up of PPP loans by Black-owned firms can be explained by whether they had less access to accountants. To study the role of access to accountants in facilitating PPP take-up, we use data on Florida CPA licenses.[30] We construct two measures of CPA access. First, we check whether we can match each firm's registered agent to the CPA license data on CPA firms and individuals. A registered agent is a business or individual designated to receive notice of legal or administrative actions affecting the firm. A firm's registered agent may be one of its officers or a third party such as a lawyer or CPA. Black-owned businesses that use accountants as registered agents are more likely to have access to an accountant to help with their PPP applications and likely have greater access to the documents they needed.

To identify registered agents who are accountants, we match registered agents based on ZIP code and name to both the individual and firm CPA license data sets. We consider a registered agent to be a CPA or CPA firm if the registered agent matches to either data set. We also tag registered agents as CPAs if the name includes words like CPA, accounting, audit, or tax. According to this algorithm, about 3% of restaurants in the sample have a registered agent that is a CPA.

Because firms can use CPAs without appointing a CPA as their registered agent, we also construct a measure of the number of local CPAs who could potentially help firms with their PPP applications or were available previously to help firms with their accounting needs. Many CPAs work for large public accounting firms or for non-accounting employers. These CPAs are unlikely to help small businesses apply for PPP loans. To focus on CPAs who are likely to help small businesses, we limit our sample of CPAs to individual CPA licenses that we can link to firm CPA licenses.[31]

Finally, we identify the race of CPAs. Given the evidence of ethnic homophily in various settings,[32] it is reasonable to expect that Black-owned firms will be more likely to use a CPA if there is a larger supply of Black CPAs nearby. Indeed, among all Florida firms with CPA registered agents, Black-owned firms are 20 times more likely to have a Black CPA as their registered agent than are white-owned firms in the same ZIP code. To identify the race of CPAs in our data, we match CPAs to Florida voter registration data using the same algorithms as for identifying the race and ethnicity of firm owners. For each ZIP code, we calculate the number of all CPAs and of Black CPAs who were active as of February 15, 2020. The average restaurant in our sample is located in a ZIP code with 0.24 Black CPAs and 9.37 total CPAs who are likely to work with small businesses.

Table 9 reports the results. We interact all demographic dummies with the CPA variables, but for brevity we report only the interactions for Black-owned restaurants. The interactions for other groups are generally not statistically significant. In column 1, the dependent variable is the overall take-up of PPP loans. The coefficient on *CPA agent* is 4.8 percentage points with a t-statistic of 1.55. The coefficient on *Black* × *CPA agent* is 23.3 percentage points, statistically significant at 5%. This result indicates that for restaurants with a CPA registered agent, there is no disparity in the take-up of PPP loans.[33] While we cannot rule out selection effects, this result is consistent with the idea that Black-owned firms with CPA registered agents have greater access to CPAs to help with PPP applications and thus overcome the impediments to receiving a PPP loan, thereby reducing disparities in PPP take-up.

In columns 2 and 3, we investigate whether the use of CPA registered agents affects the take-up of bank and nonbank PPP loans. While the coefficients on *Black* × *CPA agent* are not statistically significant in these regressions, the estimated coefficient is much larger in the nonbank regression in column 3 than in the bank regression in column 2. One potential explanation of this finding is that CPAs are particularly valuable in helping with nonbank applications because nonbanks have fewer resources to help firms with their applications. Finally, column 4 shows that *Black* × *CPA agent* has a negative effect on the take-up of EIDL loans. The magnitude of the coefficient is very similar to the PPP regression in column 1, suggesting that Black-owned firms are substituting between PPP and EIDL loans.

[28] In the simple univariate setting, the estimated effect of bank home mortgages is equal to the difference in the take-up of PPP loans by firms with and without owners that have bank home mortgages. Since our algorithm is unlikely to generate false positives, the estimated mean for the sample of firm owners with bank home mortgages, $\hat{\mu}_{\text{bank mtg}}$, is an unbiased estimate of the true value, $\mu_{\text{bank mtg}}$. The estimated mean for the sample of firm owners tagged as not having bank home mortgages is biased upwards however because this sample includes some firm owners that do in fact have bank home mortgages. These firm owners with bank home mortgages are highly likely to get PPP loans, but we treat them as if they did not have bank home mortgages. Specifically, the estimated mean for firms tagged as having no bank home mortgages, $\hat{\mu}_{\text{no bank mtg}}$, is a weighted average of the true value and the mean for firms with bank home mortgages $\hat{\mu}_{\text{no bank mtg}} = \frac{0.36}{0.68} \times \mu_{\text{no bank mtg}} + \frac{0.32}{0.68} \times \mu_{\text{bank mtg}}$. This says that out of the 68% of firms classified as having no bank mortgage, only $\frac{0.36}{0.68}$ or a bit more than half, do not actually have a bank mortgage. The fact that we overestimate the take-up of PPP loans by firms without bank mortgages means that we underestimate the effect of having a bank mortgage. Specifically, the estimated difference $\hat{\delta} = \hat{\mu}_{\text{bank mtg}} - \hat{\mu}_{\text{no bank mtg}} = \frac{0.36}{0.68}\delta$, or about half of the true difference, δ.

[29] The estimated coefficient on nonbank mortgages may also be biased towards zero if we underestimate the share of firms with nonbank mortgages. But this is unlikely to affect the estimated coefficient on *Black* because controlling for firm characteristics, there is no disparity between Black- and white-owned restaurants in the share of firms with nonbank mortgages.

[30] http://www.myfloridalicense.com/DBPR/certified-public-accounting/public-records/.

[31] Specifically, we match individual and firm licenses based on the business address. Some individual licenses match to multiple firm licenses that share the same street address, but that have different suite numbers. We consider such individuals to be linked to a firm even though we cannot determine the exact firm.

[32] The closest evidence that we are aware of comes from Frame et al. (2023), who find that within a given branch office, minority home mortgage loan applicants are almost 50% more likely to be handled by minority loan officers.

[33] The sum of the coefficients on Black and Black × CPA agent is positive but not statistically different from zero.

Table 9
Access to CPAs.

	PPP			
	All	Bank	Nonbank	EIDL
	(1)	(2)	(3)	(4)
Black	−0.153***	−0.199***	0.046**	0.048
	(0.035)	(0.034)	(0.022)	(0.036)
Hispanic	−0.081***	−0.088***	0.007	0.028
	(0.017)	(0.018)	(0.009)	(0.018)
Asian	0.010	−0.084***	0.094***	0.019
	(0.020)	(0.022)	(0.014)	(0.022)
Other	0.016	−0.024	0.040**	0.022
	(0.032)	(0.033)	(0.020)	(0.035)
Female	−0.017	−0.020	0.004	0.010
	(0.013)	(0.013)	(0.007)	(0.014)
CPA agent	0.048	0.017	0.032	0.030
	(0.031)	(0.034)	(0.020)	(0.038)
Black × CPA agent	0.233**	0.030	0.202	−0.258**
	(0.113)	(0.162)	(0.133)	(0.123)
Black × # CPAs	0.000	0.000	0.000	−0.000
	(0.001)	(0.001)	(0.000)	(0.001)
Black × # Black CPAs	0.081***	0.038	0.043**	−0.011
	(0.028)	(0.028)	(0.022)	(0.029)
R^2	0.183	0.194	0.098	0.093
ZIP FEs	✓	✓	✓	✓
Controls	✓	✓	✓	✓

This table reports the results of linear probability model regressions of receiving emergency loans on minority owner dummies interacted with (1) a dummy for whether the firm's registered agent is a CPA and (2) variables capturing the number of CPAs in the firm's ZIP code. For brevity only the interactions with Black and the direct effect of CPA registered agent are reported. Robust standard errors are reported. *, **, and *** indicate statistical significance at 10%, 5%, and 1%. $N = 12,812$.

Turning to the number of CPAs in the firm's ZIP code, we find a large positive and statistically significant coefficient on *Black × # Black CPAs* of 0.081, but no effect of total CPAs on PPP take-up by Black-owned firms. This is consistent with the view that Black-owned firms are more likely to use CPAs if there are more Black CPAs nearby. The magnitude of the coefficient implies that having access to one Black CPA in the same ZIP code is associated with a 8.1 percentage point smaller disparity in PPP take-up. Columns 2 and 3 show that this result is driven mostly by the take-up of nonbank loans, which is consistent with our finding on the effect of CPA registered agents.

Internet Appendix Table IA3 shows that these results are robust to controlling for the interactions of minority-owner dummies with other ZIP code characteristics such as the number of bank branches per capita, median household income, and Black population share. The estimated coefficient on the interaction of *Black* with the number of Black CPAs is positive and statistically significant, while the interactions of *Black* with the other ZIP characteristics are not statistically significant. These results suggest that it is access to Black CPAs rather than other correlated ZIP code characteristics that explains higher PPP take-up.

Overall, our results suggest that access to professional help in preparing PPP applications mitigates disparities in PPP take-up. Our results also point to an important limitation of fintech lenders in expanding access and reducing racial disparities in access to credit. Because fintechs have limited capacity to help firms fix any deficiencies in their applications, fintechs work best when minority-owned businesses have access to professional help.

5.3. Racial bias

In this section, we look at the effect of racial bias on PPP borrowing. Specifically, we ask whether minority-owned restaurants are less likely to receive PPP loans in locations with more racial bias in the white population. We use data on explicit and implicit racial bias collected by Project Implicit.[34] Project Implicit offers a number of online implicit association tests that test takers anywhere in the world can take to measure their implicit social attitudes. We use the results of the Race test, also known as the Black/White test, which measures an individual's implicit preference for white over Black people.[35] The test also asks subjects about their explicit preferences on a seven-point scale, where 1 is "I strongly prefer African Americans to European Americans", 4 is "I like European Americans and African Americans equally", and 7 is "I strongly prefer European Americans to African Americans".

Implicit bias has been used extensively in social sciences and has been shown to be associated with discriminatory behavior. Glover et al. (2017) show that the implicit bias of grocery store managers affects the performance of minority cashiers. On days when they are supervised by biased managers, minority cashiers are more likely to be absent from work and take longer to scan items and check out customers. Hehman et al. (2018) show that local implicit racial bias is associated with disproportionate use of lethal force against Black people.

We calculate the county-level averages across white respondents taking the test between 2008–2019. The median county has 347 valid responses. Glades County has the fewest responses, just 14. Fig. 1 plots county-level averages of explicit and implicit bias. Larger values and darker colors indicate more substantial bias. The correlation between explicit and implicit bias across all counties is 0.33. The correlation almost doubles to 0.58 when we limit the sample to 42 counties with at least 100 responses.

Table 10 reports the results of linear probability models for PPP loans and bank PPP loans in Panel A, and nonbank PPP loans and EIDL loans in Panel B. The main variables of interest are the minority restaurant-owner dummies and their interactions with county-level explicit and implicit bias. Odd-numbered columns report the results for explicit bias, while even-numbered columns report the results for implicit bias. County-level bias measures are standardized to have zero mean and unit standard deviation so the interaction coefficients capture the effect of a one standard deviation increase in racial bias.[36] The direct effect of bias in a county is absorbed by the ZIP code fixed effects included throughout. To account for the likelihood that bias in counties with low response rates may be measured with more noise, we weigh observations by the county response rate—the ratio of white test takers to the number of white residents in the county. Equal-weighted regressions generate slightly weaker results.[37] Standard errors are clustered by county to match the level at which we measure the key explanatory variable. All regressions include the full set of controls used in Table 9: restaurant characteristics, UCC loans, and measures of access to CPAs interacted with minority-owned dummies.

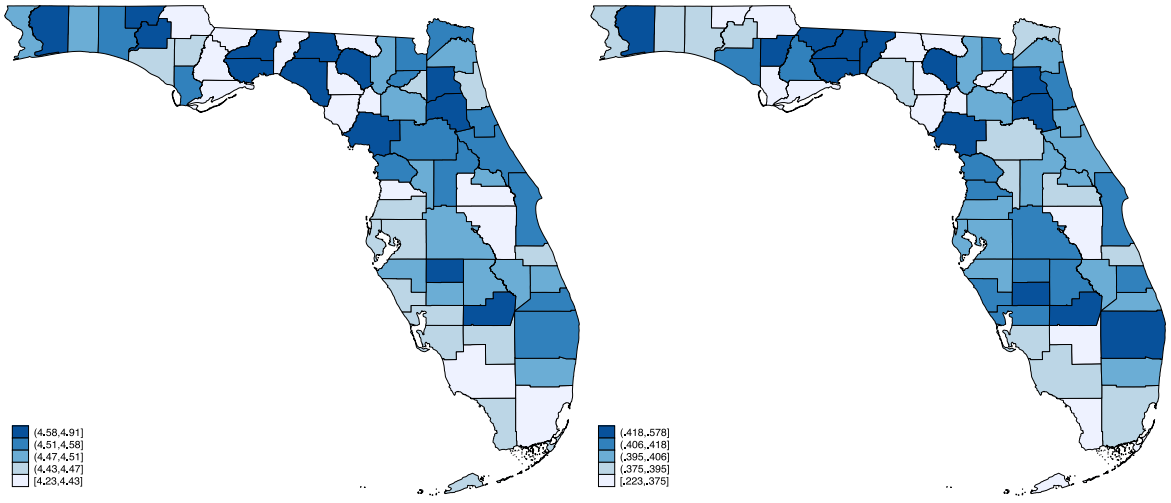
In columns 1 and 2, we examine whether minority-owned restaurants located in more racially biased counties are less likely to receive PPP loans. In both regressions, the interaction term between *Black* and *Bias* is negative and sizable in magnitude but imprecisely estimated and therefore not statistically significant. When we turn to the all industries sample in Section 6, we will find interaction terms with similar magnitude but more precisely estimated. These results will

[34] https://osf.io/y9hiq/.

[35] Project Implicit also offers a test of implicit attitudes towards Asian Americans, which one could potentially use to ask whether businesses owned by Asian Americans are less likely to receive bank PPP loans in counties with stronger anti-Asian bias. The Asian Implicit Association Test has far fewer responses, however. Over the 2014–2019 period during which a test taker's county is included in the data, no results are available for 10 out of 67 Florida counties. The median county has only 17 responses by white test takers.

[36] We standardize within the sample of Florida counties so that the interpretation of the coefficients is based on the cross-county variation within Florida.

[37] We find qualitatively similar results when weighting by the number of test-takers in the county. The number of test takers, however, is strongly correlated with county population.



(a) Explicit bias (b) Implicit bias

Fig. 1. Explicit and implicit racial bias across Florida counties. This figure reports county-level averages of explicit and implicit racial bias of white respondents to the Race Implicit Association Test. Larger values indicate stronger bias against African Americans. Explicit bias is measured on a scale from 1 to 7, with subjects explicitly stating whether they "strongly prefer African Americans to European Americans" (1) or they "strongly prefer European Americans to African Americans" (7). Implicit bias is the score on the implicit association test. Tests taken during 2008–2019 are included. The median county has 347 respondents. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 10
Racial bias.

	PPP		Bank PPP		Nonbank PPP		EIDL	
	Explicit (1)	Implicit (2)	Explicit (3)	Implicit (4)	Explicit (5)	Implicit (6)	Explicit (7)	Implicit (8)
Black	−0.226***	−0.186***	−0.287***	−0.221***	0.062	0.035	0.068*	0.039
	(0.079)	(0.034)	(0.052)	(0.032)	(0.039)	(0.023)	(0.035)	(0.036)
Hispanic	−0.077***	−0.069***	−0.052	−0.055*	−0.025**	−0.014	0.046	0.056
	(0.027)	(0.023)	(0.031)	(0.032)	(0.011)	(0.013)	(0.041)	(0.043)
Asian	−0.006	−0.004	−0.101***	−0.087***	0.095***	0.083***	0.004	0.004
	(0.019)	(0.024)	(0.024)	(0.028)	(0.016)	(0.018)	(0.025)	(0.022)
Other	−0.023	0.028	−0.035	−0.011	0.012	0.040	0.007	0.021
	(0.066)	(0.064)	(0.056)	(0.049)	(0.023)	(0.031)	(0.051)	(0.042)
Female	−0.013	−0.023**	−0.016	−0.026**	0.003	0.003	0.005	0.012
	(0.010)	(0.009)	(0.012)	(0.010)	(0.012)	(0.009)	(0.014)	(0.012)
Black × Bias	−0.105	−0.077	−0.176**	−0.191*	0.071	0.114*	0.074***	0.121***
	(0.135)	(0.145)	(0.076)	(0.106)	(0.065)	(0.059)	(0.025)	(0.036)
Hispanic × Bias	−0.030	−0.042	−0.002	−0.010	−0.028	−0.032	−0.022	−0.001
	(0.024)	(0.028)	(0.029)	(0.036)	(0.018)	(0.025)	(0.040)	(0.055)
Asian × Bias	−0.012	0.032	−0.048	−0.026	0.037	0.058	−0.002	−0.001
	(0.049)	(0.066)	(0.059)	(0.090)	(0.028)	(0.041)	(0.047)	(0.066)
Other × Bias	−0.142**	−0.199**	−0.065	−0.129	−0.078**	−0.069	−0.041	−0.024
	(0.065)	(0.082)	(0.067)	(0.078)	(0.036)	(0.071)	(0.063)	(0.105)
Female × Bias	0.028*	0.032	0.028*	0.028	−0.000	0.004	−0.017	0.005
	(0.016)	(0.021)	(0.015)	(0.018)	(0.012)	(0.014)	(0.022)	(0.030)
R^2	0.187	0.187	0.193	0.192	0.092	0.091	0.086	0.086
ZIP FEs	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓

This table reports the results of linear probability model regressions of receiving different types of emergency loans on the restaurant owner's race interacted with explicit and implicit racial bias against Black people:

$$Emergency\ loan_{c,f,r,z} = \alpha_z + \beta \cdot Minority_f + \delta \cdot Female_f + \theta \cdot Minority_f \times Racial\ bias_c + \gamma' X_{f,r} + \varepsilon_{f,c,r,z},$$

where c indexes counties, f indexes firms, r indexes restaurants, and z indexes ZIP codes. Regressions are weighted by the county-level response rate to the Race Implicit Association Test. The response rate is the number of white respondents taking the test during the 2008–2019 period divided by the number of white residents in the county in 2019. The sample construction is summarized in Tables A.2 and 1. Controls include all restaurant characteristics included in the regressions of Table 9. Standard errors are adjusted for clustering by county. *, **, and *** indicate statistical significance at 10%, 5%, and 1%. Continuous variables are standardized so that their coefficients represent the effect of a one standard deviation change. $N = 12,812$.

provide stronger evidence that Black-owned businesses are even less likely to receive PPP loans in more racially biased counties.

In columns 3 and 4, we examine the effect of racial bias on the probability of receiving a PPP loan from a bank. In both regressions, the interaction between *Black* and *Bias* is negative and statistically significant. The magnitude of the effect is very large. A one standard deviation increase in racial bias is associated with a 17.6–19.1 percentage point reduction in the probability that a Black-owned restaurant receives a bank PPP loan, depending on whether we use explicit or implicit bias as the measure. Given that Black-owned restaurants are 28.7 (22.1) percentage points less likely to get a bank PPP loan in a county with average explicit (implicit) bias, this implies that Black-owned restaurants are 46.3 (41.2) percentage points less likely to receive bank PPP funding in counties with explicit (implicit) bias one standard deviation above the mean. The interactions between racial bias and the other minority restaurant-owner dummies are also negative, but smaller in magnitude and generally not statistically significant. It makes sense that although bias towards Black people may be correlated with bias against other minority groups, its strongest effect is on the likelihood that Black-owned restaurants receive bank PPP loans.

In columns 5 and 6, we present results for nonbank PPP loans. Here the coefficient on the interaction between *Black* and *Bias* is 7.1–11.4 percentage points and the opposite of the sign for bank PPP loans. It is statistically significant at 10% for implicit but not explicit bias. These results provide suggestive evidence that Black-owned restaurants are more likely to substitute from bank to nonbank PPP loans in counties with greater racial bias. Finally, in columns 7 and 8, the dependent variable is whether the firm receives an EIDL loan. As with nonbank PPP loans, we find a positive interaction between *Black* and *Bias*, suggesting more substitution to EIDL loans in more biased counties.

One may be concerned that racial bias is correlated with other county-level characteristics in a way that may explain differences in borrowing and drive the results in Table 10. Given the small number (66) of counties, it may be difficult to disentangle the effect of bias from other county characteristics. We therefore postpone doing so until we examine a broader sample of firms across four states.

6. Analysis of other industries and states

Since restaurants were hit especially hard by the pandemic, one may wonder whether our results apply to other industries. Because we cannot identify the population of potential borrowers in other industries, we restrict our analysis to firms receiving other forms of government support that had similar eligibility criteria as the PPP. Specifically, we examine whether there are racial disparities in the take-up of PPP loans among the broader sample of firms that received EIDL Advance grants, just as we did in Table 5 for restaurants. One advantage of this approach is that because we are no longer limited to restaurants, we can expand our sample to include firms in other states with voter registration data that include self-reported racial and ethnic identity and are available to researchers at a modest cost. These states are Georgia, Louisiana, and North Carolina.[38]

To create this broader sample, we start by matching EIDL Advance recipients to Florida, Georgia, Louisiana, and North Carolina corporate records, which we access through OpenCorporates, an open database of corporations around the world. We limit the resulting sample to for-profit corporations and LLCs that were active as of February 15, 2020. We next match PPP borrowers to these corporate records and use this link to determine which EIDL Advance recipients also received PPP loans.

Because neither EIDL Advance nor OpenCorporates data include industry classification, we match firms to the Dun & Bradstreet database and use the firm's primary SIC code in Dun & Bradstreet. Firms with SIC code 9999, non-classifiable establishments, are excluded from the analysis. We estimate regressions with ZIP code cross two-digit SIC industry fixed effects to compare minority- and white-owned firms that are in the same industry and same ZIP code; this corresponds to the approach in our analysis of Florida restaurants.

We match each firm's officers, directors, and agents to voter registration data using the same algorithm we use for our restaurants sample. However, we make two adjustments to our method of classifying firms based on the racial and ethnic identity of their officers. First, while Florida corporate records always report both registered agents who are designated to receive correspondence from the state and officers/directors who are authorized to act on behalf of the firm, not all other states do the same. In particular some states, including Georgia and North Carolina, report the registered agents but not necessarily officers of LLCs. With some firms using professional registered agents like CPAs and lawyers, we do not want to classify firms based on the racial identity of such professional agents. We therefore exclude individuals who are listed as registered agents of five or more firms.[39] Second, in contrast to Florida corporate records, other states in our sample are less likely to list officers in order of seniority. Thus, instead of using the racial identity of the first listed officer we can match to voter registration, we set minority indicators equal to one if any of the matched officers report their racial or ethnic identity.

Table 11 reports the results, with PPP take-up in columns 1–2, bank PPP take-up in columns 3–4, and nonbank PPP take-up in columns 5–6. Regression specifications control for ZIP-SIC2 fixed effects, thus enabling us to compare PPP take-up by firms in the same industry and ZIP code. We include the log of firm age, indicators for corporations (versus LLCs), and indicators for the implied number of employees based on the size of the EIDL Advance grant, although because of space limitations we do not report the coefficients of these variables. We also control for the interaction of other minority indicators with racial bias. To make sure that the results on the interaction between minority dummies and racial bias are not driven by other county and ZIP characteristics, we control for the interaction between minority indicators and (a) the log of county population, (b) the share of county population that identifies as Black, (c) the share of county population that identifies as white, (d) the gap in high school graduation rates between Black and white county residents, (e) the gap in unemployment rates between Black and white county residents, (f) the gap in household income between Black and white households, (g) number of bank branches per capita, and (h) the share of bank deposits in branches of the top 4 banks: Bank of America, Citigroup, JPMorgan Chase, and Wells Fargo. The full set of coefficients is reported in Table IA2 in the Internet Appendix. Standard errors are adjusted for clustering by county.

In column 1, the coefficient on *Black* is −0.125. This means that a Black-owned EIDL Advance recipient with the average values of all county characteristics and without any UCC borrowing relationships is 12.5 percentage points less likely to receive a PPP loan than a comparable white-owned firm. Firms with Hispanic officers are 7.7 percentage points less likely to receive PPP loans, while firms with Asian officers are 1.7 percentage points more likely to receive PPP loans. Although the interpretation of the coefficients on the minority indicators is slightly more nuanced in Table 11, these results are broadly similar to the results in Table 5 for the sample of restaurants participating in the EIDL Advance program.

We find a strong negative interaction between *Black* and measures of explicit and implicit racial bias. In a county that is one standard deviation above the mean of implicit racial bias, Black-owned firms are a further 4.0 percentage points less likely to receive PPP loans. These findings are driven by disparities in the take-up of bank PPP loans, which we demonstrate in columns 3–4. Column 4 shows that in a county that is one standard deviation above the mean of implicit racial bias, Black-owned firms are a further 7.9 percentage points less likely to receive bank PPP loans. In columns 5–6, the dependent variable

[38] Alabama and South Carolina also report racial and ethnic identity, but South Carolina data is not available to nonresidents of South Carolina, and Alabama data costs $37,000.

[39] Specifically, we tabulate the number of firms a given voter is associated with as an agent, organizer, or incorporator. If a voter is associated with at least five firms, we exclude this voter from the classification of these firms.

Table 11
Analysis of other industries and states.

	PPP		Bank PPP		Nonbank PPP	
	Explicit (1)	Implicit (2)	Explicit (3)	Implicit (4)	Explicit (5)	Implicit (6)
Black	−0.125***	−0.124***	−0.143***	−0.140***	0.018***	0.017***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.004)	(0.003)
Asian	0.017***	0.016***	0.003	0.003	0.014***	0.013***
	(0.005)	(0.005)	(0.006)	(0.006)	(0.004)	(0.004)
Hispanic	−0.077***	−0.077***	−0.080***	−0.080***	0.003	0.003
	(0.005)	(0.005)	(0.006)	(0.006)	(0.004)	(0.004)
Other	−0.012**	−0.012**	−0.013**	−0.013**	0.001	0.001
	(0.006)	(0.006)	(0.006)	(0.006)	(0.003)	(0.003)
Female	0.004	0.004*	0.004	0.004	0.000	0.000
	(0.002)	(0.002)	(0.003)	(0.003)	(0.002)	(0.002)
Black × Bias	−0.031**	−0.040***	−0.061***	−0.079***	0.030***	0.039***
	(0.013)	(0.015)	(0.017)	(0.019)	(0.009)	(0.011)
Bank UCC	0.080***	0.080***	0.108***	0.108***	−0.028***	−0.028***
	(0.008)	(0.008)	(0.008)	(0.008)	(0.003)	(0.003)
Nonbank UCC	0.046***	0.046***	0.025***	0.025***	0.020***	0.020***
	(0.008)	(0.008)	(0.008)	(0.008)	(0.005)	(0.005)
Black × Bank UCC loan	0.066***	0.066***	0.068***	0.068***	−0.001	−0.001
	(0.014)	(0.014)	(0.015)	(0.015)	(0.010)	(0.010)
Black × Nonbank UCC loan	0.025**	0.025**	−0.002	−0.002	0.028***	0.028***
	(0.012)	(0.012)	(0.013)	(0.013)	(0.007)	(0.007)
R^2	0.390	0.390	0.392	0.392	0.190	0.190
ZIP-SIC2 FEs	✓	✓	✓	✓	✓	✓
Firm chars	✓	✓	✓	✓	✓	✓
Minority × Bias	✓	✓	✓	✓	✓	✓
Minority × County/ZIP chars	✓	✓	✓	✓	✓	✓
Minority × UCC	✓	✓	✓	✓	✓	✓

This table reports the results of linear probability model regressions of receiving PPP loans conditional on receiving an EIDL Advance grant. The sample consists of all firms in Florida, Georgia, Louisiana, and North Carolina that receive EIDL Advance and that can be matched to OpenCorporates, voter registration, and Dun & Bradstreet. Information on UCC loans is from MailingLists. Firm characteristics include the log of firm age (years between incorporation date and February 15, 2020), a dummy for corporation (versus LLC), dummies for the implied number of employees based on the size of the EIDL Advance grant, and dummies for having active UCC loans from bank and nonbank lenders. County characteristics include the log of county population, the share of county population that identifies as white, and gaps between Black and white county residents in high school graduation rate, unemployment rate, and household income. County characteristics are from the 2019 ACS. ZIP code characteristics are the number of bank branches per capita and the share of ZIP code deposits in branches of the top 4 banks: Bank of America, Citigroup, JPMorgan Chase, and Wells Fargo. Standard errors are adjusted for clustering by county. *, **, and *** indicate statistical significance at 10%, 5%, and 1%. N = 275,572.

is whether the firm received a nonbank PPP loan. These columns show evidence of substitution to nonbanks, especially in counties with stronger racial bias. For example, in column 6 the interaction between *Black* and implicit bias is 0.039, which is highly significant statistically and economically.

Regression specifications in Table 11 also include interactions of *Black* with bank and nonbank UCC loan dummies. The interaction with bank UCC is around 0.066–0.068 in PPP and bank PPP regressions in columns 1–2 and 3–4, and is zero in the nonbank PPP regressions in columns 5–6. These results indicate that the disparity in the take-up of PPP loans is significantly smaller for firms with pre-existing bank UCC borrowing relationships.[40]

Overall, the results in Table 11 show that there are large racial disparities in the larger sample of EIDL Advance recipients across all industries and four states. Black-owned firms are significantly less likely to receive PPP loans, especially in more biased counties. Pre-existing bank borrowing relationships and substitution to nonbanks help mitigate but do not eliminate disparities in PPP borrowing.

7. Conclusion

We have documented racial disparities in the Paycheck Protection Program and examined their causes. About 60% of disparities for Black- and Hispanic-owned restaurants can be explained by a combination of location and restaurant characteristics. Pre-existing borrowing relationships do little to explain disparities on average. Black-owned restaurants are still 9.1 percentage points less likely to receive PPP funding than similar white-owned restaurants. For Hispanic-owned restaurants the difference is a more modest 5.6 percentage points, and for Asian-owned restaurants and female-owned restaurants there is no appreciable difference in PPP take-up.

Disparities in PPP borrowing are driven by disparities in PPP borrowing from banks. Black-owned restaurants are 16.8 percentage points less likely than white-owned restaurants to borrow from banks, while Hispanic-owned restaurants are 5.8 percentage points less likely, and Asian-owned restaurants are 6.1 percentage points less likely. Nonbanks — largely fintechs — make up for a portion of these disparities by lending at greater rates to Black-owned restaurants (7.6 percentage points), and they make up the entire difference for Asian-owned restaurants. Hispanic-owned restaurants borrow from nonbanks at the same rate as white-owned restaurants.

The disparity in bank PPP borrowing for Black-owned restaurants appears to be exacerbated by racial bias. Restaurants located in counties where more white people express implicit or explicit biases towards Black people are less likely to receive PPP loans from banks. They are also more likely to substitute to nonbank PPP loans and EIDL loans. We see this basic pattern across a wide range of industries in Florida, Georgia, Louisiana, and North Carolina. Racial bias may have affected banks' administration of PPP loans, or it may have reflected a legacy of bias that deterred Black restaurant owners from applying for PPP loans from banks. Our methodology cannot tell apart these two explanations, but our findings suggest that online, less personalized applications can help mitigate bias.[41] Both fintechs and the SBA used

[40] In untabulated results, we included triple interactions between *Black*, bank relationships, and racial bias to see whether bank relationships have a larger effect on the take-up of PPP loans by Black-owned firms in areas with greater racial bias. Our data do not appear to have enough statistical power to test for such a triple interaction — the estimates are not statistically significant and have large standard errors.

[41] In a more recent paper, Chernenko et al. (2024) use survey data to show that racial bias reduces both the probability that Black-owned businesses apply to banks and the probability that they are approved conditional on applying.

Table A.1
Variable definitions.

Variable	Definition
	Firm characteristics
Number of seats	Number of seats reported on the restaurant's license.
Firm age	Years between February 15, 2020 and the first filing date in Florida corporate records.
Bank/nonbank UCC loan	Indicator variable for whether the firm had any active UCC filings as of February 15, 2020 with banks/nonbanks as the secured party. We match debtors in Florida UCC filings to Florida corporate records. We match secured parties to the list of financial institutions maintained by the National Information Center and to Capital IQ. UCC3 filings are used to track continuations, terminations, and changes in debtor and secured parties.
UCC bank's PPP intensity	Bank's PPP intensity is the ratio of (a) PPP loans to Florida-based borrowers to (b) UCC loans to Florida-based borrowers that were active as of February 15, 2020. For firms with UCC loans from multiple banks, we calculate the average intensity across banks. For firms without bank UCC loans, intensity is set to zero.
Number of reviews	Number of Yelp reviews as of February 2020.
Average rating	Average Yelp rating as of February 2020. Set to zero for restaurants without any reviews. We separately include an indicator variable for no reviews.
Page views	Average number of monthly Yelp profile views between March 2019 and February 2020.
Number of photos	Average number of photos posted to the restaurant's Yelp profile each month between March 2019 and February 2020.
Accepts credit cards	Indicator variable equal to one if the restaurant accepts credit cards according to its Yelp profile. Set to one for restaurants with missing information. We separately include an indicator variable for missing credit card information. Measured as of March 11, 2020.
Employees	Number of employees is the value of EIDL Advance grant divided by 1000.
CPA agent	Indicator variable equal to one if the firm's registered agent in Florida corporate records is a CPA. We match registered agents to lists of individual and firm CPA licenses maintained by Florida's Department of Business & Professional Regulation http://www.myfloridalicense.com/DBPR/certified-public-accounting/public-records/
	County characteristics
Explicit racial bias	Response to the question "Which statement best describes you? 1 'I strongly prefer African American to European Americans', 2 'I moderately prefer African Americans to European Americans' ...4 'I like European Americans and African Americans equally' ...7 'I strongly prefer European Americans to African Americans'." (variable name `att7`) We calculate county-level average across all white respondents taking the test between 2008 and 2019. County-level racial bias is standardized to have zero mean and standard deviation equal to one.
Implicit racial bias	Overall score on the Race Implicit Association Test (variable name `D_biep.White_Good_all`). We calculate county-level average across all white respondents taking the test between 2008 and 2019. County-level racial bias is standardized to have zero mean and standard deviation equal to one.
	ZIP code characteristics
Bank branches per capita	Number of bank branches in the ZIP code divided by population. Number of bank branches in the ZIP code is from the June 2019 Summary of Deposits. Estimate of the 2019 ZIP Code Tabulation Area (ZCTA) population (variable name `B02001_001E`) is from the 2019 American Community Survey.
Deposit share of the top 4 banks	The value of deposits in branches of Bank of America, Citigroup, JPMorgan Chase, and Wells Fargo relative to total deposits June 2019 Summary of Deposits. Estimate of the 2019 ZIP Code Tabulation Area (ZCTA) population (variable name `B02001_001E`) is from the 2019 American Community Survey.
White population share	Estimate of white alone population (variable name `B02001_002E`) divided by estimate of total population (variable name `B02001_001E`) from the 2019 American Community Survey.
Median household income	Median household income in the past 12 months (variable name `S1901_C01_012E`) from the 2019 American Community Survey.
Population	Estimate of total population (variable name `B02001_001E`) from the 2019 American Community Survey.
Black CPAs	Number of Black CPAs working for accounting firms. We identify the race of CPA license holders by matching them to Florida voter registration data. To focus on CPAs who are likely to help small businesses, we limit the sample of individual CPA licenses to those whose address matches the address of one of the firms in the firm CPA license data. We take the maximum of the number of Black CPAs in the ZIP of the restaurant itself and in the ZIP of the firm.

an online application process with relatively less personal interaction than the application process used by many banks.

At the same time, our results point to an important limitation of fintech lenders in expanding access and reducing racial disparities in access to credit. Because fintechs have limited capacity to help firms fix any deficiencies in their applications, fintechs work best when minority-owned businesses have access to professional help. Having just one more Black CPA in a firm's ZIP code is associated with a 4.3 percentage point higher take-up of nonbank PPP loans by Black-owned firms.

Our findings also point to potential issues with the design of the PPP that may have inadvertently affected the access of minority-owned businesses to PPP loans. While our focus has been on understanding racial disparities after controlling for differences in restaurant attributes — including proximity to bank branches, age, size, and borrowing history — these differences led to substantially lower PPP take-up by minority-owned restaurants. Indeed, these differences explain 60% of the disparity in borrowing by Black- and Hispanic-owned restaurants. While it is possible that some of the attributes of minority-owned restaurants may have lowered their demand for PPP loans, it is also possible that there were impediments to getting PPP loans. Thus, quite apart from issues of racial bias, a program that effectively favored larger businesses in close proximity to banks or with strong borrowing relationships with banks, even if it did so unintentionally, was less successful than it might have been in serving the emergency funding needs of minority-owned businesses.

Table A.2
Florida restaurants sample.

Step	*N* affected	*N* remaining
0. Seating licenses approved before Feb 16, 2020		41,266
1. Drop out-of-state license holders	5,685	35,581
3. Drop hotel restaurants	3,218	32,363
4. Drop franchise restaurants	6,932	25,431
5. Drop restaurants operated by municipalities and JVs	96	25,335
6. Match to Florida corporate records	2,390	22,945
7. Drop firms incorporated after February 15, 2020	810	22,135
8. Drop non-profits, out-of-state, and publicly-traded firms	1,679	20,456
9. Match owner to voter registration	6,073	14,383
10. Match to Yelp	1,571	12,812

This table summarizes the sample selection criteria and the number of restaurants at each step in the construction of the sample. Details of the matching algorithm used in the sample construction are described in the Internet Appendix.

Table A.3
Nonbank lenders.

Lender	*N*	%
CROSS RIVER BANK	282	19.0
KABBAGE	202	13.6
WEBBANK	196	13.2
CELTIC BANK CORP	187	12.6
READYCAP LENDING LLC	136	9.2
ITRIA VENTURES LLC	116	7.8
FOUNTAINHEAD SBF LLC	47	3.2
SQUARE CAPITAL LLC	40	2.7
HARVEST SMALL BUSINESS FINANCE LLC	40	2.7
NEWTEK SMALL BUSINESS FINANCE	29	2.0
INTUIT FINANCING	24	1.6
BSD CAPITAL LLC	20	1.3
BENWORTH CAPITAL	20	1.3
MBE CAPITAL PARTNERS	15	1.0
FC MARKETPLACE LLC	15	1.0
FUNDBOX	12	0.8
DREAMSPRING	12	0.8
CAPITAL PLUS FINANCIAL LLC	12	0.8
CENTERSTONE SBA LENDING	11	0.7
PRESTAMOS CDFI LLC	9	0.6
SUNSHINE STATE ECONOMIC DEVELOPMENT CORP	9	0.6
LIBERTY SBF HOLDINGS LLC	7	0.5
ENTERPRISE CENTER CAPITAL CORP	7	0.5
AMUR EQUIPMENT FINANCE	6	0.4
A10CAPITAL LLC	6	0.4
ASCENDUS	5	0.3
BLACK BUSINESS INVESTMENT FUND	3	0.2
TIMEPAYMENT CORP	3	0.2
CRF SMALL BUSINESS LOAN COMPANY LLC	3	0.2
LIFTFUND	2	0.1
OPPORTUNITY FUND COMMUNITY DEVELOPMENT	2	0.1
INDEPENDENT DEVELOPMENT SERVICES CORP	1	0.1
HOPE ENTERPRISE CORP	1	0.1
WORLD TRADE FINANCE	1	0.1
AMERICAN LENDING CENTER	1	0.1

This table lists nonbank lenders extending PPP loans to restaurants in our data. Loans held by banks that serve as partner banks to nonbank lenders that originate PPP loans are considered nonbank loans. Cross River Bank, Web Bank and Celtic Bank are the three largest partner banks.

CRediT authorship contribution statement

Sergey Chernenko: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **David Scharfstein:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

First author received research support from Purdue University's Blake Family Fund for Ethics, Leadership and Governance. I have no other potential conflicts to disclose.

Second author received research support from Harvard Business School. I was previously on the Board of Directors of M&T Bank Corporation, which made loans through the Paycheck Protection Program. I have no other potential conflicts to disclose.

Data availability

Racial Disparities in the Paycheck Protection Program (Original data) (Mendeley Data).

Appendix A

See Tables A.1–A.3.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2024.103911.

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